



CHALMERS
UNIVERSITY OF TECHNOLOGY

MMS236 - Aircraft Design

Lecture 3

Carlos Xisto

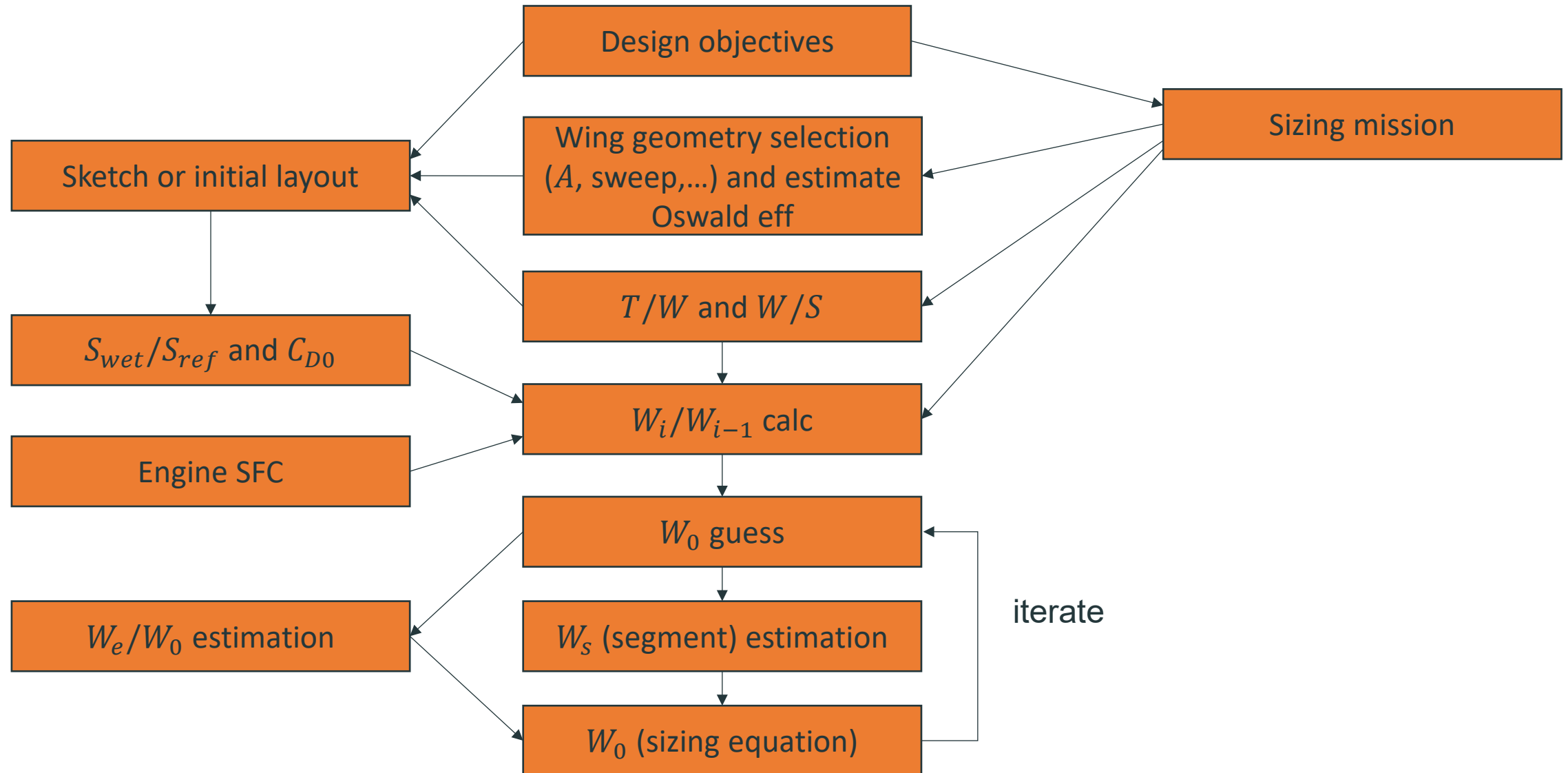
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Department of Mechanics and Maritime Sciences



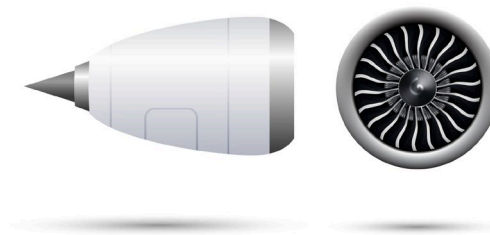
Conceptual layout

Sizing (continuation). Aerodynamic and Structural Considerations. Crew station, passengers and payload, landing gear a subsystems

Initial sizing
(better method)



- ✈ Steps through the mission to find fuel burn for each segment.
- ✈ Allows for payload drop.
- ✈ Can be used to size while considering a "fixed" (existing) engine or "rubber" (new) engine.



Remember the sizing equation?

$$\star W_0 = W_{crew} + W_{payload} + W_{dropped\ payload} + W_f + W_e$$

Or

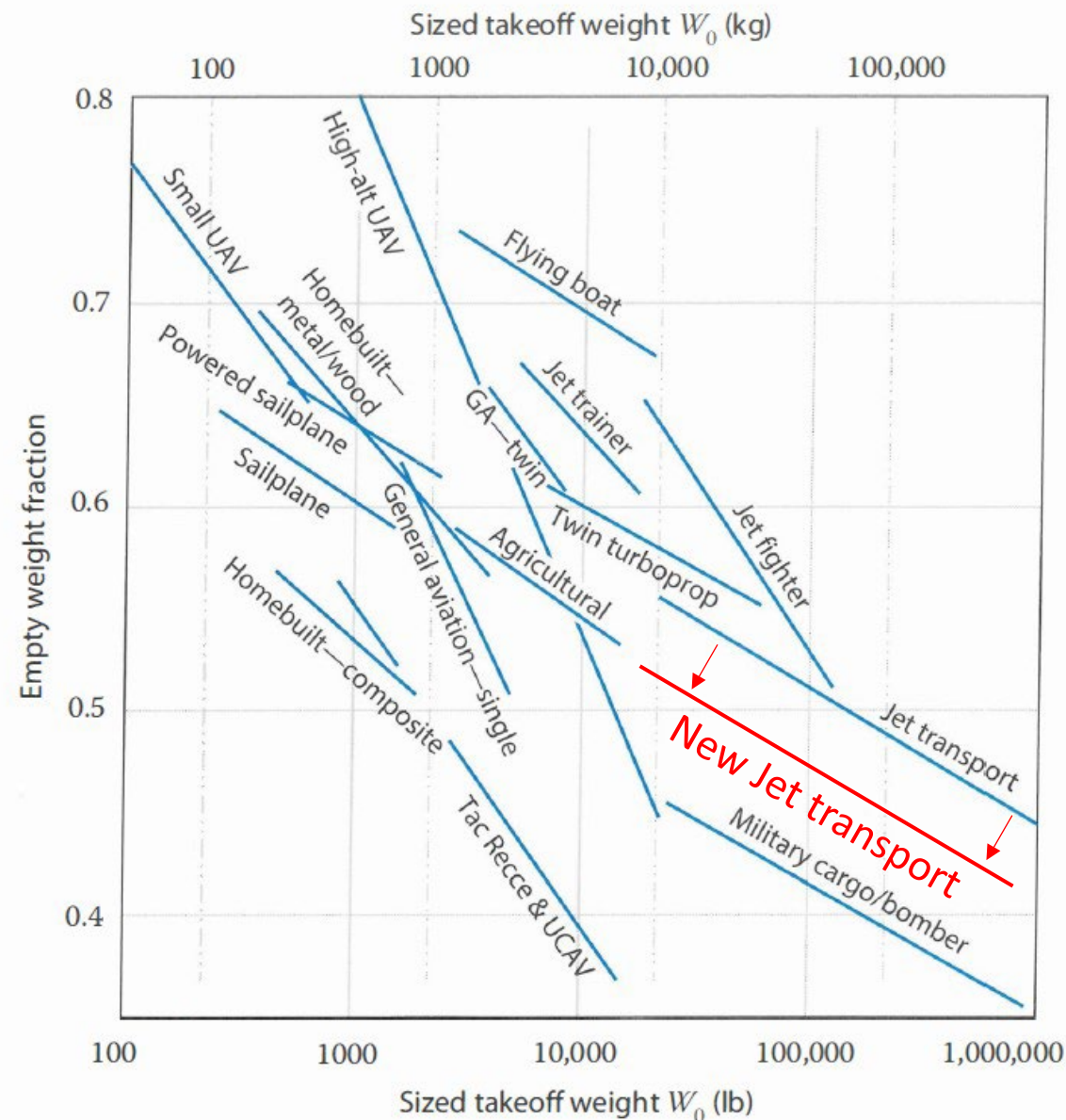
$$\star W_0 = W_{crew} + W_{payload} + W_{dropped\ payload} + W_f + \frac{W_e}{W_0} W_0$$

$\star \frac{W_e}{W_0}$ - predicted using statistical methods at this point - *we have not designed the aircraft!*

Empty weight fraction

Best approach

- ✈ Gather as much data as possible on similar aircraft (the more recent the better).
- ✈ Create your own $\frac{W_e}{W_0}$ vs. W_0 correlation.
 - remember that $\frac{W_e}{W_0}$ normally goes down with increasing W_0 .
- ✈ Apply fudge factors as needed.



Alternative (more time consuming)

- ✈ Use more independent variables (A, W/S, T/W, M) Tab. 6.1 and 6.2 in Raymer.

$\frac{W_e}{W_0} = \left[a + b W_0^{c1} A^{c2} \left(\frac{T}{W_0} \right)^{c3} \left(\frac{W_0}{S} \right)^{c4} M_{\max}^{c5} \right] \cdot K_{vs}$							
fps units	<i>a</i>	<i>b</i>	C1	C2	C3	C4	C5
Jet fighter	-0.02	2.16	-0.1	0.2	0.04	-0.1	0.08
Military Cargo	0.07	1.71	-0.1	0.1	0.06	-0.1	0.05
Jet Transport	0.32	0.66	-0.13	0.30	0.06	-0.05	0.05

K_{vs} - variable sweep constant: 1.04 for variable sweep wings; 1.0 otherwise.

Note: Don't forget to convert from lbs to kg.

✦ We now assume the possibility of payload drop

✦ The segment fuel burn is:

$$W_{fi} = \left(1 - \frac{W_i}{W_{i-1}}\right) W_{i-1} \quad \frac{W_i}{W_{i-1}} - \text{mission segment weight fractions}$$

1. Start by W_0 (guess) = W_1
2. Apply mission-segment weight fraction to calculate W_{f1}
3. $W_2 = W_1 - W_{f1} - W_{dropp}$ (if any)
4. Repeat

✦ The mission fuel burn is (including 5% contingency and 1% trapped fuel):

$$W_f = 1.06 \sum_1^i W_{fi}$$

$$\left(\frac{W_i}{W_{i-1}}\right)_{\text{cruise}} = \exp\left(\frac{-R \cdot (SFC)_{\text{cruise}} \cdot g_0}{V \left(\frac{L}{D}\right)_{\text{cruise}}}\right)$$

Where in cruise Lift = Weight, hence:

$$\left(\frac{L}{D}\right)_{\text{cruise}} = \frac{C_L}{C_D} = \frac{C_L}{\underbrace{C_{D_0}}_{\text{Parasite}} + \underbrace{C_{Di}}_{\text{Induced}} + \underbrace{C_{D_{\text{Lift-viscous}}}}_{\text{Lift dependent viscous}} + \underbrace{C_{D_{\text{wave}}}}_{\text{Compressibility drag}}}$$

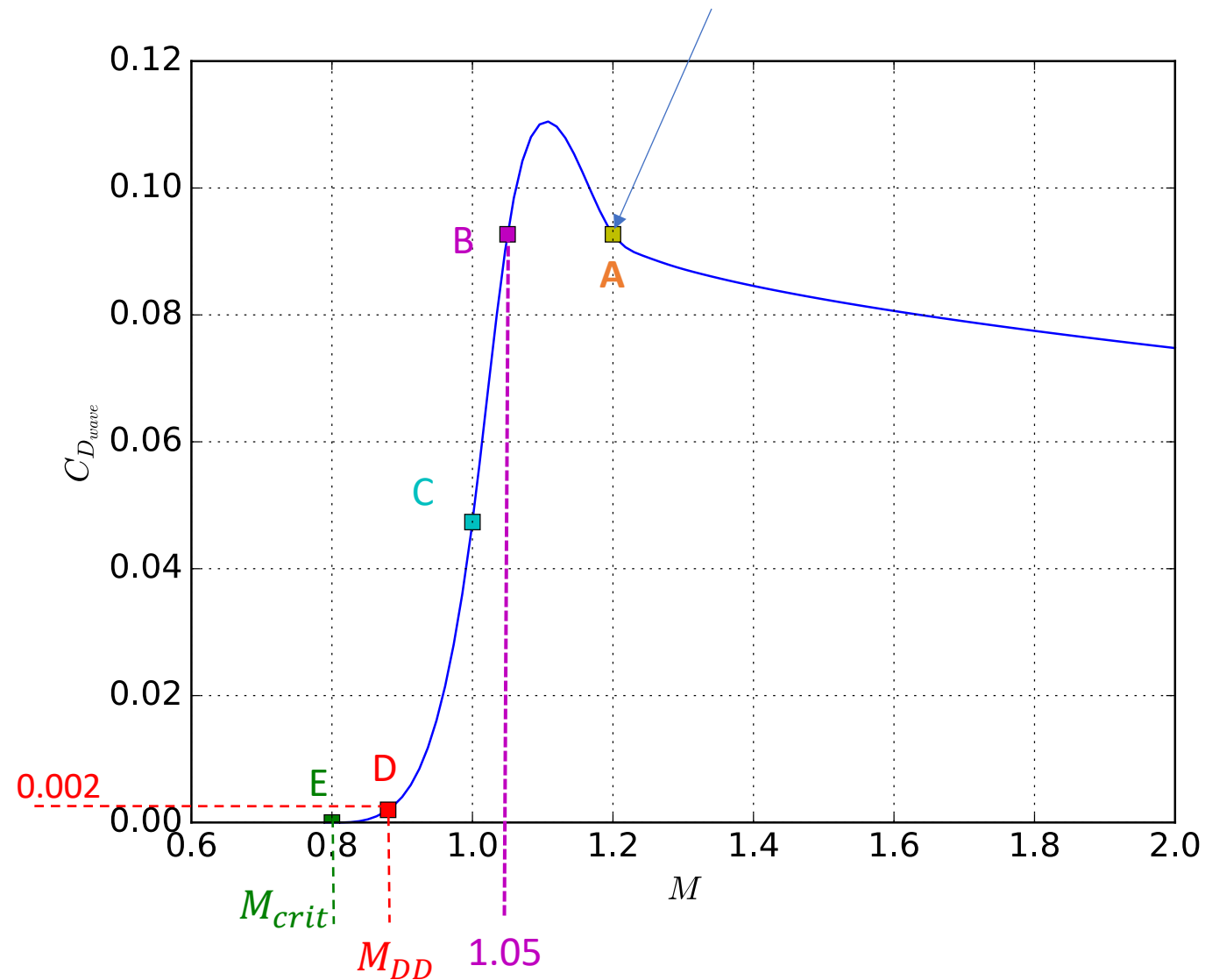
Lift dependent

$$\left(\frac{L}{D}\right)_{\text{cruise}} = \frac{C_L}{C_D} = \frac{C_L}{C_{D_0} + C_{Di} + C_{D_{\text{Lift-viscous}}} + C_{D_{\text{wave}}}}$$

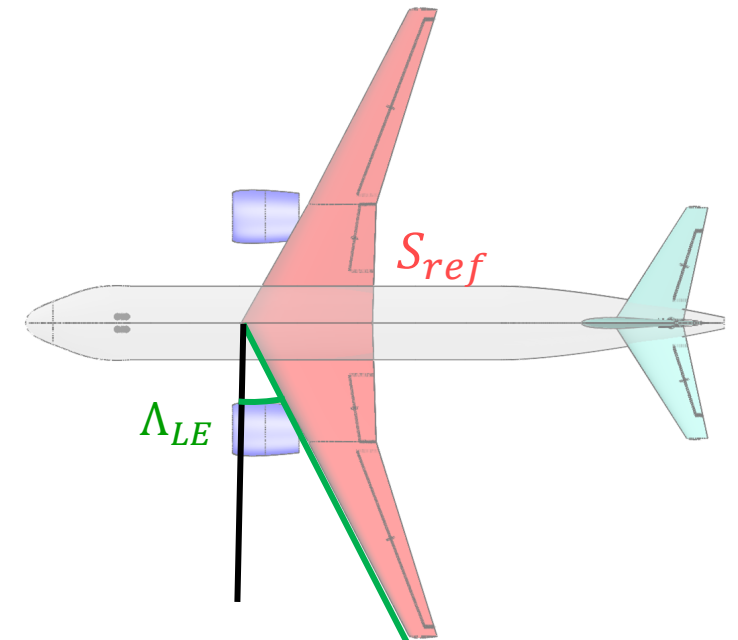
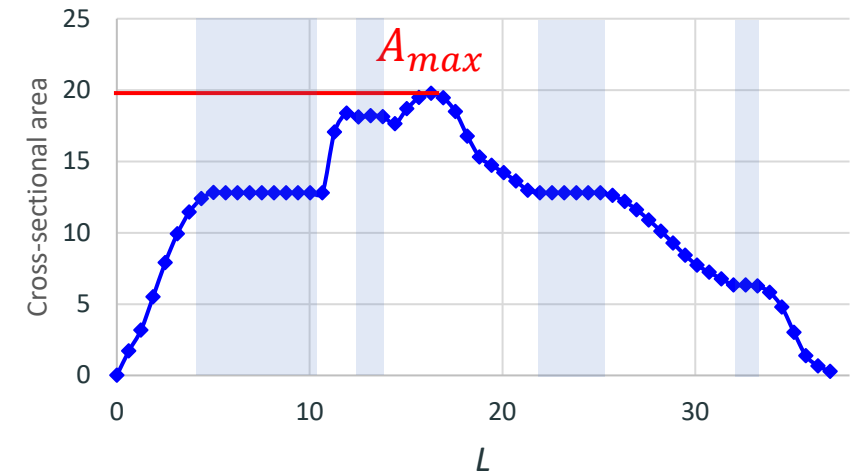
Where:

- C_{D_0} : Parasite drag, typically 0.015 for transport aircraft (use VSP)
- $C_{Di} = \frac{1}{\pi A e} C_L^2$, where A is the aspect ratio and e is the Oswald efficiency
- $C_{D_{\text{Lift-viscous}}} = K_0 C_{D_0} C_L^2$, where $K_0 = 0.38$ is a factor determined using flight data.
- $C_{D_{\text{wave}}}$, is the drag due to compressibility effects usually determined from a wave drag plot

$$\left(\frac{D}{q}\right)_{wave} = E_{WD} \left[1 - 0.386(M - 1.2)^{0.57} \left(1 - \frac{\pi \Lambda_{LE}^{0.77}}{100} \right) \right] \left(\frac{D}{q}\right)_{Sears-Haack}$$



$$\left(\frac{D}{q}\right)_{Sears-Haack} = \frac{9\pi}{2} \left(\frac{A_{max}}{l} \right)^2 \quad l = L - shaded$$

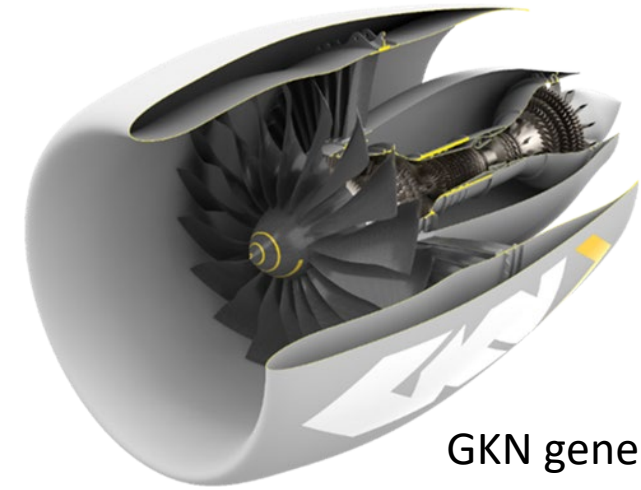


Engine:

$$\text{✈ } T = \frac{W_0}{N} \cdot g_0 \left(\frac{T}{W} \right)_{\text{takeoff}} \quad (\text{Newton})$$

$$\text{✈ } P = \frac{W_0}{N} \left(\frac{P}{W} \right)_{\text{takeoff}} \quad (\text{kW, } P/W \text{ given in kW/kg})$$

N – number of engines

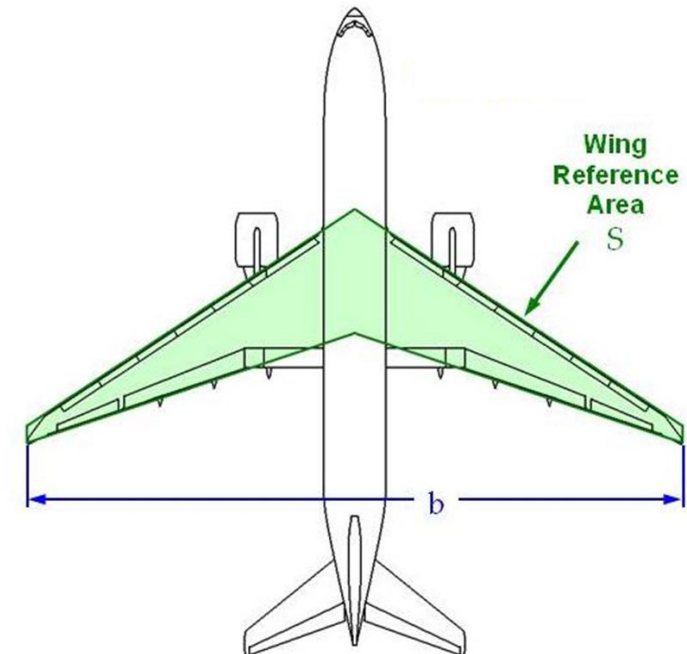


GKN generic turbofan

Wing:

$$\text{✈ } S = \frac{W_0}{\left(\frac{W_0}{S} \right)_{\text{takeoff}}} \quad (\text{reference area})$$

✈ Use eqs. provided in lecture 2 to layout the wing.



- ✈ Existing engines are very often re-used.
- ✈ T/W is no longer constant.
- ✈ Achieve either mission range or performance – **not both**



V2500 engine on a 2015 Embraer KC390



V2500 engine on a 1992 A320ceo

Constant performance

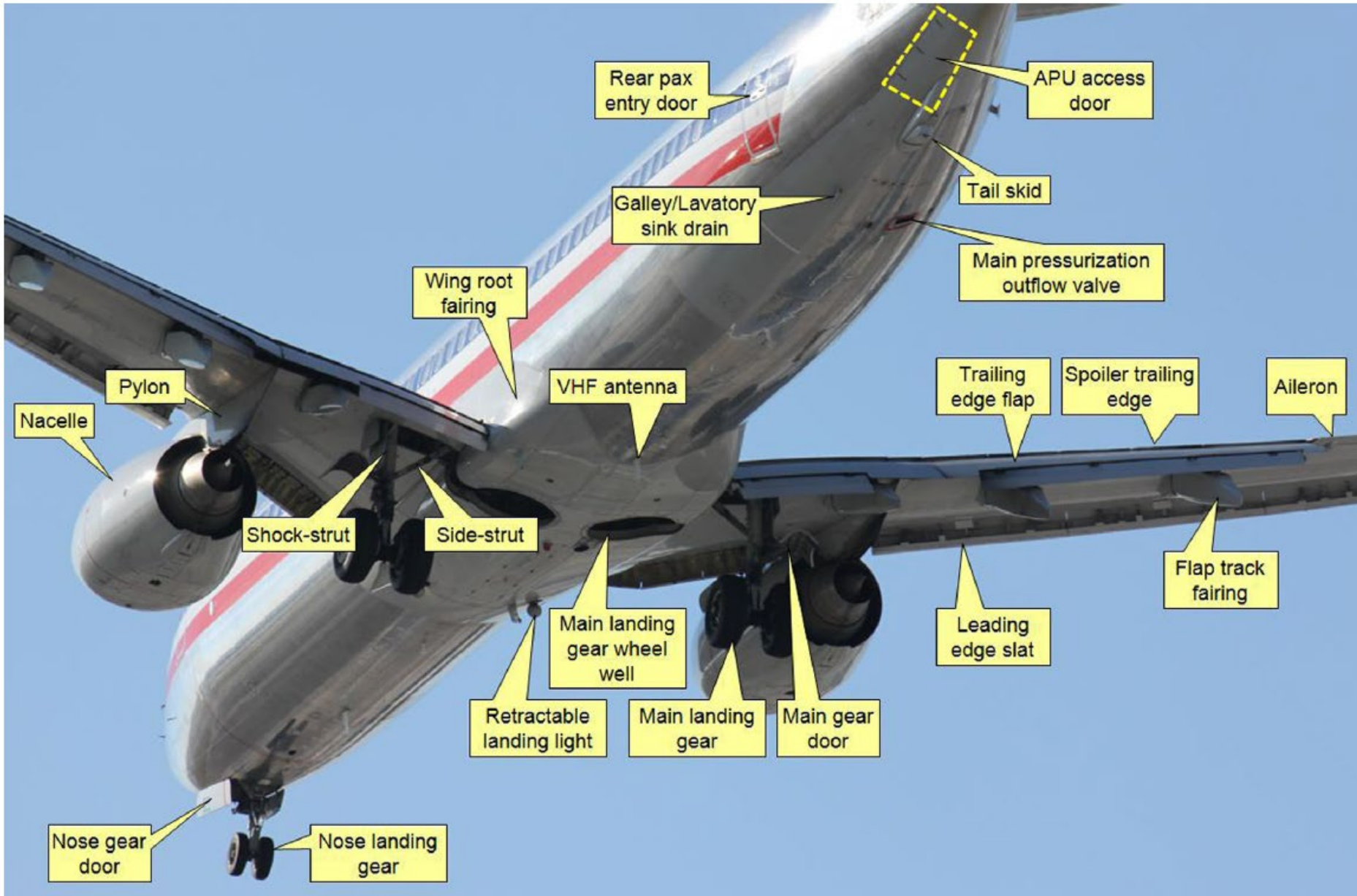
- ✦ Determine T/W – constant
- ✦ Calculate $W_0 = \frac{T \cdot N}{\left(\frac{T}{W}\right)_{\text{takeoff}}}$
- ✦ Iterate on sizing mission range until W_0 is satisfied

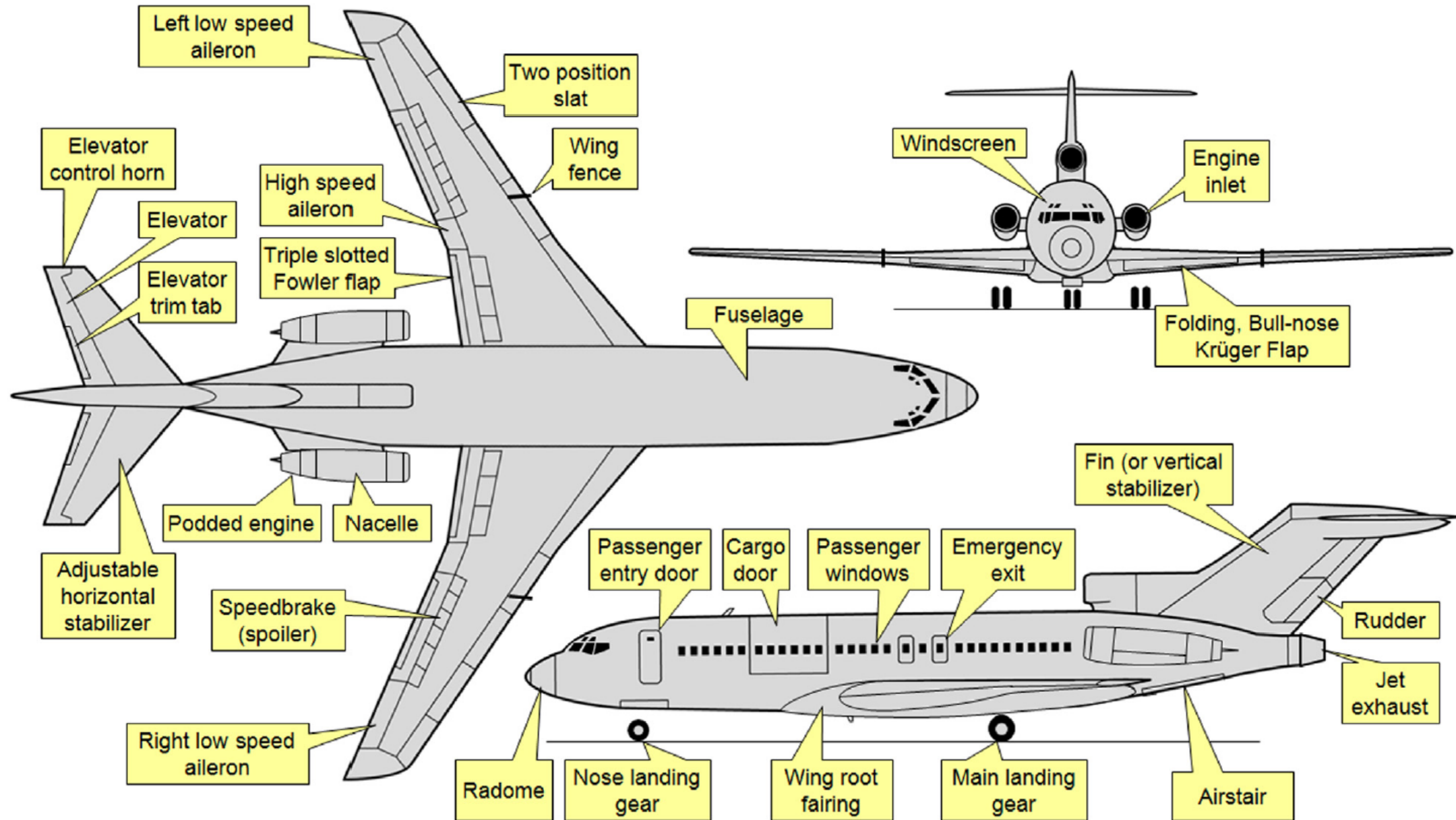
Constant range

- T/W will change as W_0 is computed to meet the required range
- The performance needs to be calculated after establishing T/W

Conceptual Layout

- ✈ The requirements, mission definition, and regulations lead the way. => How far? How fast? How high? How heavy? What is the airplane suppose to do? Which regulations apply?
- ✈ Past configurations with similar requirements lead to safe development paths. => Proven safety and reliability.
 - <https://www.airbus.com/en/airport-operations-and-technical-data/aircraft-characteristics>
 - https://www.boeing.com/commercial/airports/plan_manuals.page
 - [Q400](#)
- ✈ More radical solutions might lead for better performance, but are risky.
- ✈ Nice looking aircraft fly better! (matter of taste).





Make sure everything fits!

CHALMERS

De Havilland Canada Dash 8-300 Cutaway Drawing Key

- 1 Radome
- 2 Weather radar scanner
- 3 Radar mounting bulkhead
- 4 ILS glideslope aerial
- 5 Nose compartment access doors
- 6 Taxiing lamp
- 7 Radar transmitter/receiver
- 8 Transformer rectifier units
- 9 Oxygen bottles
- 10 Electrical distribution box
- 11 Battery
- 12 Nosewheel bay
- 13 Nose undercarriage leg doors
- 14 Nosewheel forks
- 15 Twin nosewheels
- 16 Ground power socket
- 17 Front pressure bulkhead
- 18 Rudder pedals
- 19 Instrument panel
- 20 Windscreen wipers
- 21 Instrument panel shroud
- 22 Windscreen panels
- 23 Overhead systems switch panel
- 24 Co-pilot's seat
- 60 Floor beam construction
- 61 Cabin window panels
- 62 Navigation system electronics equipment
- 63 Underfloor air conditioning ducting
- 64 External floodlights
- 65 Main cabin honeycomb floor panels
- 66 Seat mounting rails
- 67 Cabin wall trim panelling
- 68 Cabin roof trim and lighting panels
- 69 Port side stowage bins
- 70 Wing attachment fuselage main frames
- 71 Centre wing panel rib construction
- 72 De-icing control valve
- 73 Kevlar honeycomb wing root fairing

- 25 Observer's folding seat
- 26 Control column handwheel
- 27 Pilot's seat
- 28 Safety harness
- 29 Document case
- 30 Cockpit floor level
- 31 Underfloor control runs and air conditioning ducting
- 32 Control system access panel
- 33 Pilot head
- 34 Circuit breaker panel
- 35 Curved cockpit side window panels
- 36 Fire extinguisher
- 37 Cockpit rear bulkhead
- 38 Control cable runs
- 39 Electrical distribution panel
- 40 Cockpit roof escape hatch
- 41 Starboard side toilet compartment
- 42 Wash hand basin
- 43 VHF aerial
- 44 Starboard side type 1 emergency exit doorway
- 45 Wardrobe compartment
- 46 Aft facing pair of seats
- 47 Interphone
- 48 Cabin attendants' folding seat
- 49 External inspection light
- 50 Avionics equipment racks
- 51 Airstairs external handle
- 52 Passenger entry door/airstairs
- 53 Folding handrail
- 54 Entry lobby
- 55 Floor mounted escape lighting
- 56 Four-abreast passenger seating, 60-seat layout
- 57 Forward fuselage plug section
- 58 Overhead stowage bins
- 59 Fuselage frame and stringer construction

- 74 Engine bleed air ducting
- 75 Engine control runs
- 76 Inboard leading-edge de-icing boot
- 77 Starboard main undercarriage leg struts
- 78 Starboard engine nacelle
- 79 Pratt & Whitney Canada PW123 turboprop

- 80 Engine accessory equipment
- 81 Propeller reduction gearbox
- 82 Engine air intake
- 83 Hamilton Standard 14SF-15 four-bladed variable pitch reversible propeller
- 84 Spinner
- 85 Propeller hub pitch-change mechanism
- 86 Engine accessory equipment access panels
- 87 Twin landing lamps
- 88 Outer wing panel/joint rib
- 89 Starboard wing integral fuel tank; total fuel capacity 850 US gal (3 271 l)

- 90 Leading-edge stall strip
- 91 Wing access panels
- 92 Fuel system piping
- 93 Overwing fuel filler cap
- 94 Pneumatic leading-edge de-icing boot
- 95 Extended wing-tip section
- 96 Starboard navigation light
- 97 Wing-tip fairing
- 98 Static dischargers
- 99 Starboard aileron
- 100 Aileron trim tab
- 101 Aileron spring tab
- 102 Aileron hinge control
- 103 Outer flap screw jack
- 104 Single differential roll control spoiler, open
- 105 Flap track fairings
- 106 Starboard single-slotted flap, down position
- 107 Flap guide rail
- 108 Starboard engine exhaust duct
- 109 Exhaust shroud
- 110 Pressure refuelling connection
- 111 Main undercarriage wheel bay
- 112 Inner wing panel optional extended range fuel tank
- 113 Inboard flap segment
- 114 Trailing-edge wing root fillet
- 115 Control cable linkages
- 116 Flap hydraulic motor
- 117 Fire extinguisher bottles
- 118 Flap screwjack housing
- 119 Flap drive shaft
- 120 Port inboard single-slotted flap segment
- 121 Emergency exit window panel, port and starboard
- 122 Rear cabin window panels
- 123 Overhead passenger service units

- 134 Emergency location transmitter aerial
- 135 Fin leading-edge flush HF aerial
- 136 Tailfin construction
- 137 Rudder hydraulic actuators
- 138 Fin skin panels
- 139 Elevator cable pulley
- 140 Elevator control rods
- 141 Tailplane centre-section attachment
- 142 Upper position light
- 143 Anti-collision light
- 144 Tailplane leading-edge pneumatic de-icing boot
- 145 Starboard tailplane
- 146 Starboard elevator
- 147 Elevator trim tabs
- 148 Elevator spring tabs
- 149 Port elevator rib construction
- 150 Static dischargers
- 151 Elevator horn balance
- 152 Tailplane construction
- 153 Two-segment rudder construction
- 154 Fore rudder
- 155 Trailing rudder
- 156 Lower position light
- 157 Heat exchanger exhaust duct

- 124 Stowage bins
- 125 Starboard side service door/emergency exit
- 126 Galley unit
- 127 Baggage compartment access door
- 128 Rear fuselage plug section
- 129 Up-and-over baggage/cargo door, open
- 130 Bleed air supply duct to air conditioning system
- 131 Fin root fillet
- 132 Heat exchanger flush air intake
- 133 Emergency location transmitter
- 158 Tailcone
- 159 Ventral access hatch
- 160 Sloping fin attachment main frames
- 161 Flight data and cockpit voice recorders
- 162 Rear fuselage frame and stringer construction
- 163 Twin-pack air conditioning plant

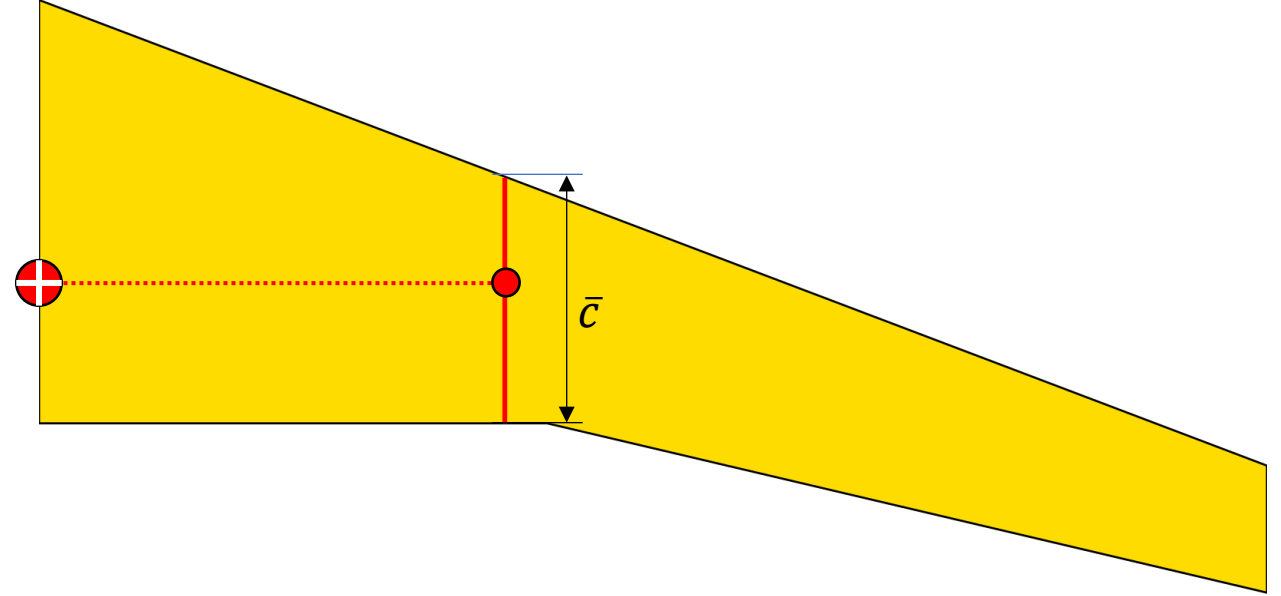
- 164 Heat exchanger
- 165 Optional auxiliary power unit (APU) on starboard side
- 166 Rear pressure bulkhead
- 167 Baggage restraint net
- 168 Baggage/cargo bay floor
- 169 Baggage door guards

- 170 Port nacelle tail fairing
- 171 Exhaust duct shroud
- 172 Port engine exhaust pipe
- 173 Honeycomb trailing-edge shroud panels
- 174 Port outer single slotted flap
- 175 Flap down position
- 176 Port differential roll control spoiler
- 177 Flap rib construction
- 178 Aileron hinge control
- 179 Aileron spring tab
- 180 Port aileron rib construction
- 181 Static dischargers
- 182 Aileron mass balance
- 183 Glassfibre wing-tip fairing
- 184 Port navigation light
- 185 Port wing-tip extension segment
- 186 Leading-edge de-icing boot
- 187 Leading-edge honeycomb core panels
- 188 Rear spar
- 189 Fuel filler cap
- 190 Wing rib construction
- 191 Port wing integral fuel tank
- 192 Front spar
- 193 Pneumatic de-icing valves
- 194 Leading-edge stall strip
- 195 Hydraulic equipment bay
- 196 Outer wing panel joint rib
- 197 Main undercarriage leg upper yoke
- 198 Mainwheel leg doors
- 199 Main undercarriage faired leg strut
- 200 Twin mainwheels
- 201 Hydraulic brake pipos

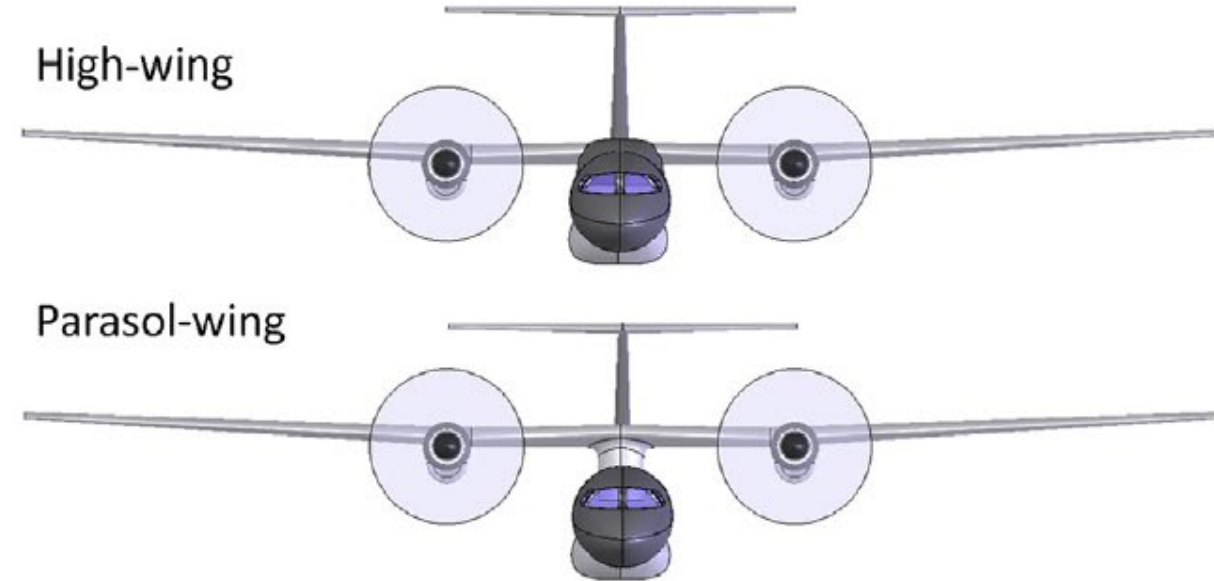
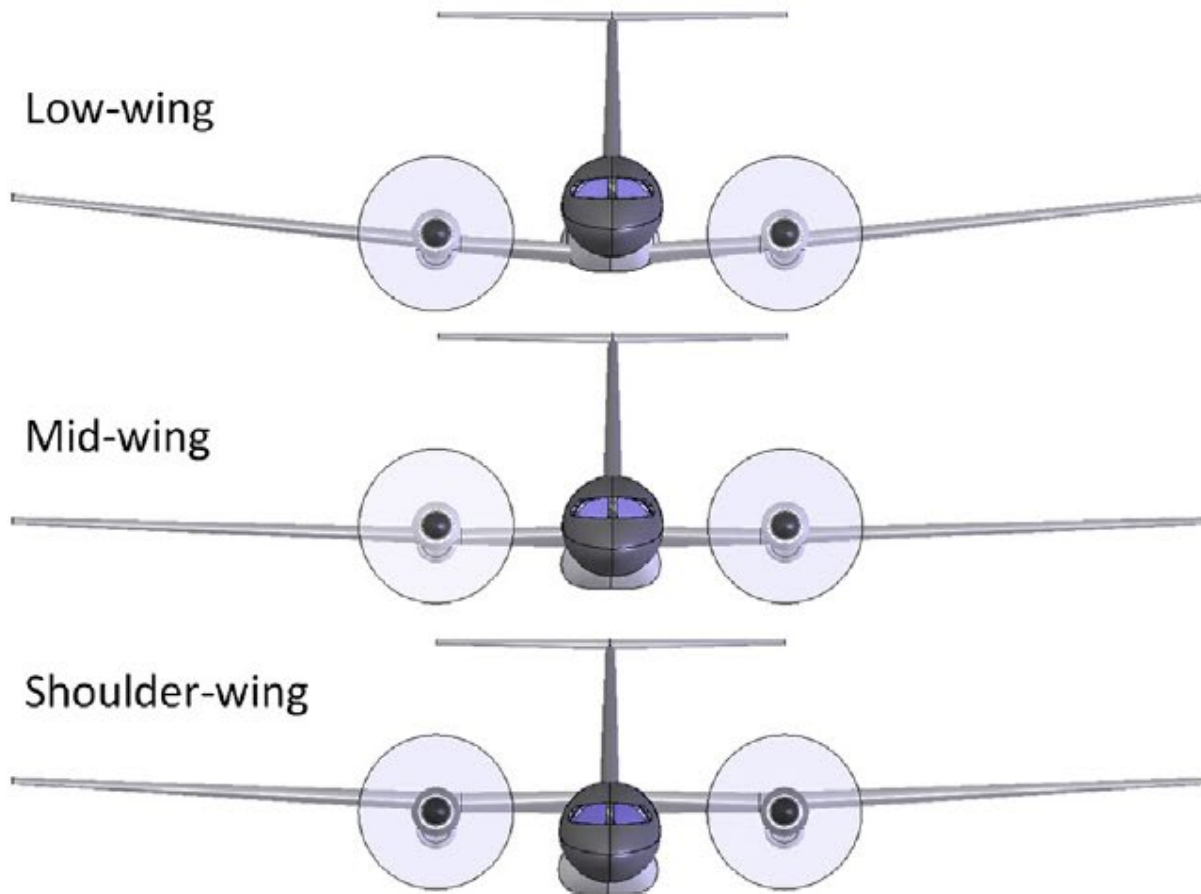
- 202 Faired forward V-strut
- 203 Main lag breaker strut
- 204 Twin landing lamps
- 205 Undercarriage mounting frame
- 206 Engine oil cooler
- 207 Engine bay firewall
- 208 Intake snow and debris ejector
- 209 Wing inspection lamp
- 210 Engine bearer struts
- 211 Intake duct
- 212 Forward engine mounting ring frame
- 213 Port propeller spinner
- 214 Intake lip de-icing

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DRAWING

- ✈ Flying wing, no tail
 - CG at 25% of $\bar{c}$ (MAC)
- ✈ Stable aircraft, with aft-tail:
 - CG at 30% of $\bar{c}$ (MAC)
- ✈ Unstable aircraft, with aft-tail:
 - CG and 40% of $\bar{c}$ (MAC)

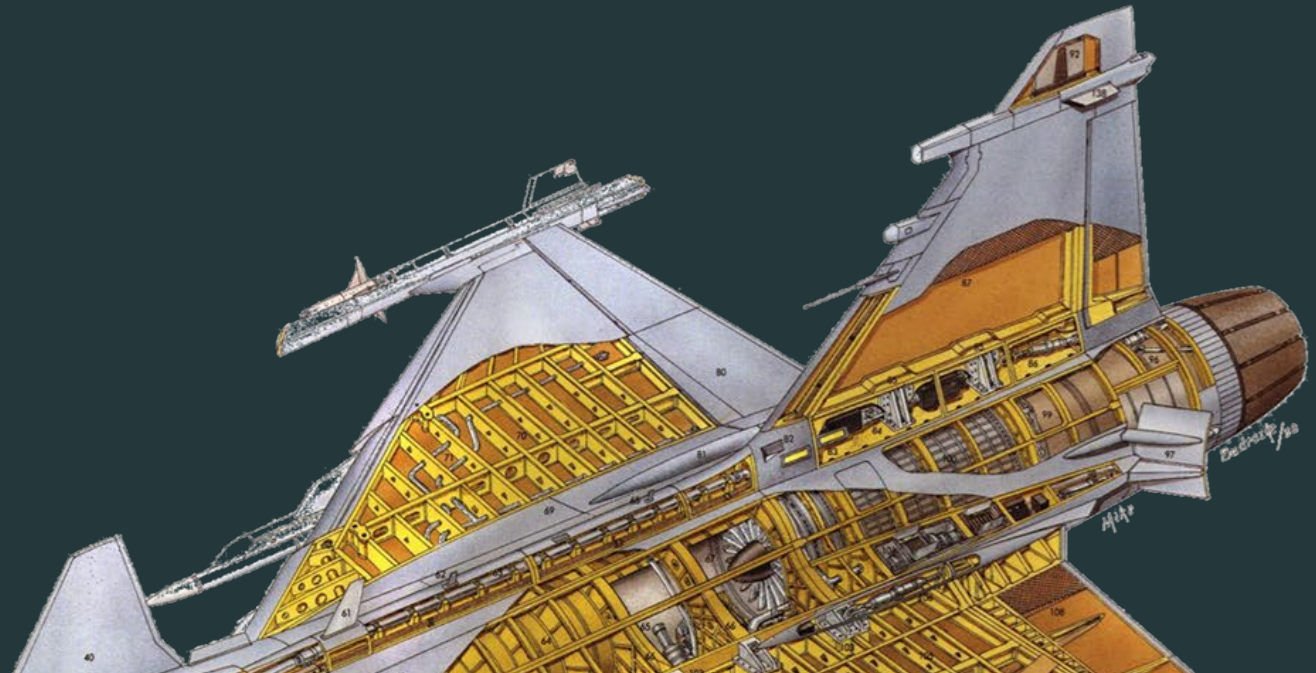


- ✈ Aircraft CG:
 - Estimated after detailed component weight break down.
 - Moves during the mission and for payload variations, cannot go beyond limits set in CG envelope.
 - Historically:
 - For front engine aircraft: 35% of fuselage length
 - Engine on wing: 40 – 50% of fuselage length
 - Aft engine: 45 – 55% of fuselage length



Source: Snorri G (2014), *General aviation aircraft design: Applied methods and procedures*, Butterworth-Heinemann, 1st Ed.

Aerodynamic and structural considerations in layout

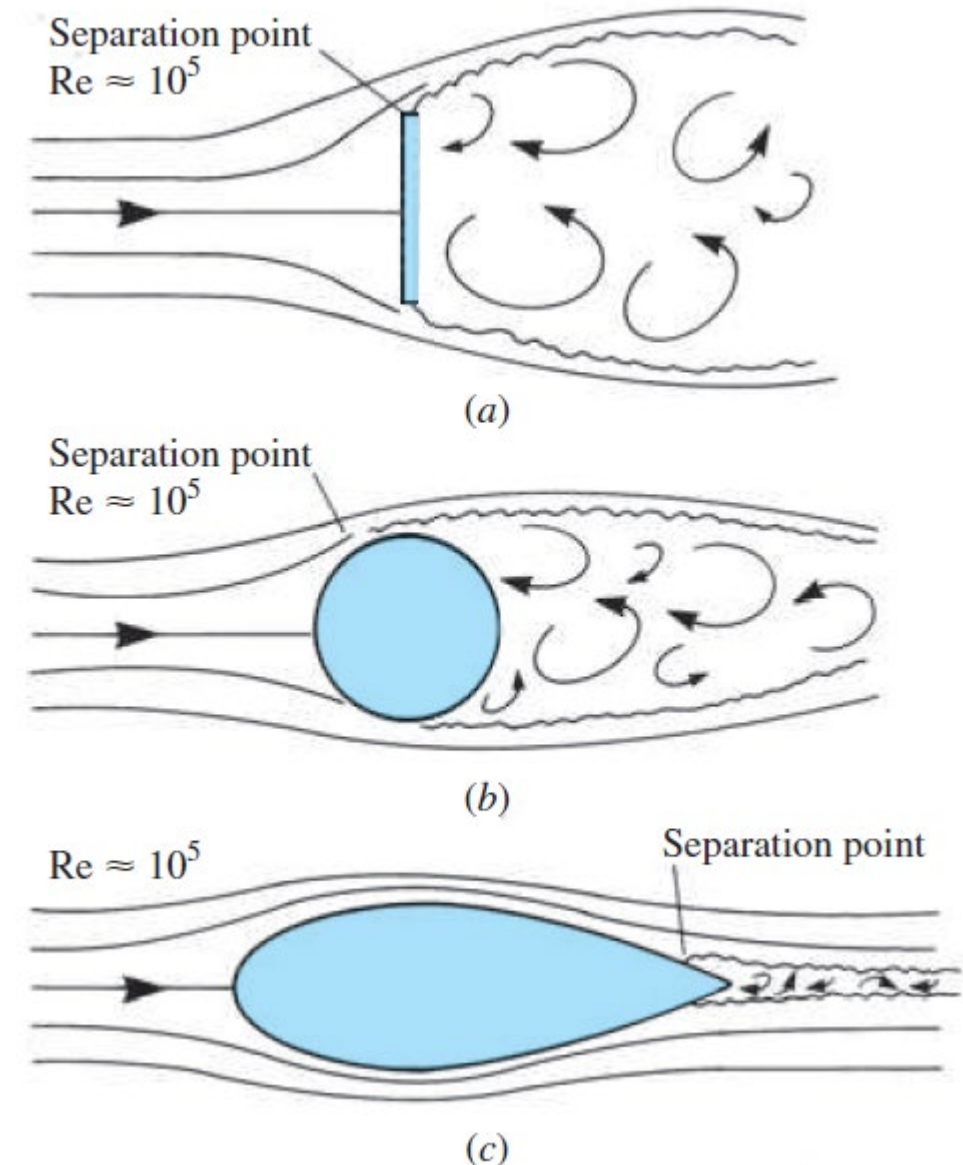


To improve aerodynamics:

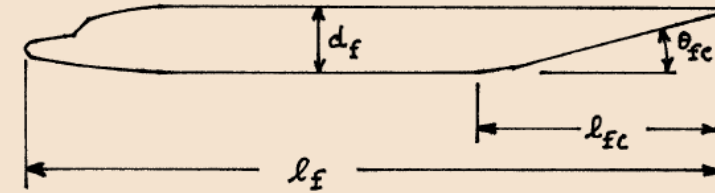
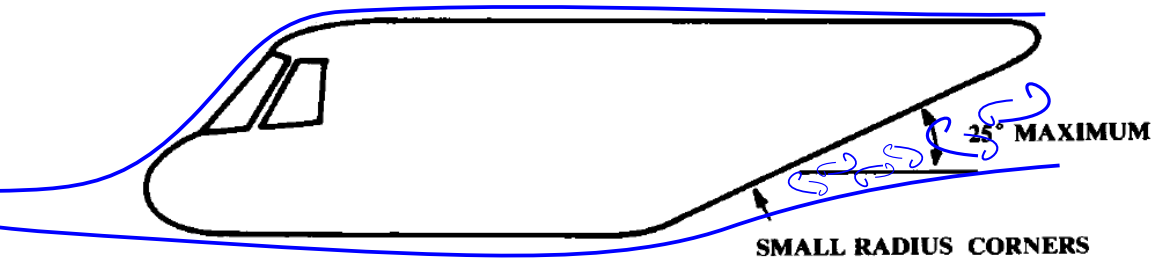
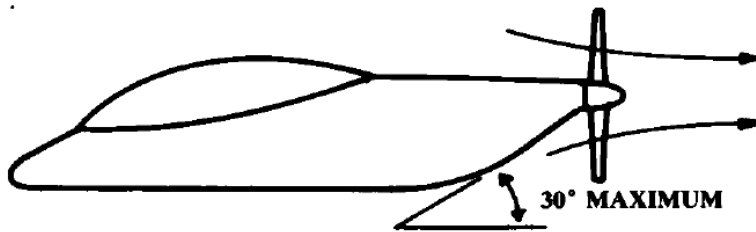
- ✈ Reduce frontal and **wetted area**.
- ✈ Clean configuration with minimum interference drag.
- ✈ Adequate fineness ratio – (6-8 for subsonic aircraft):

$$f = \frac{l_{fuse}}{d_{max}}$$

- ✈ Avoid sharp corners.
- ✈ An experienced designer can "see air".



Shaping the fuselage:



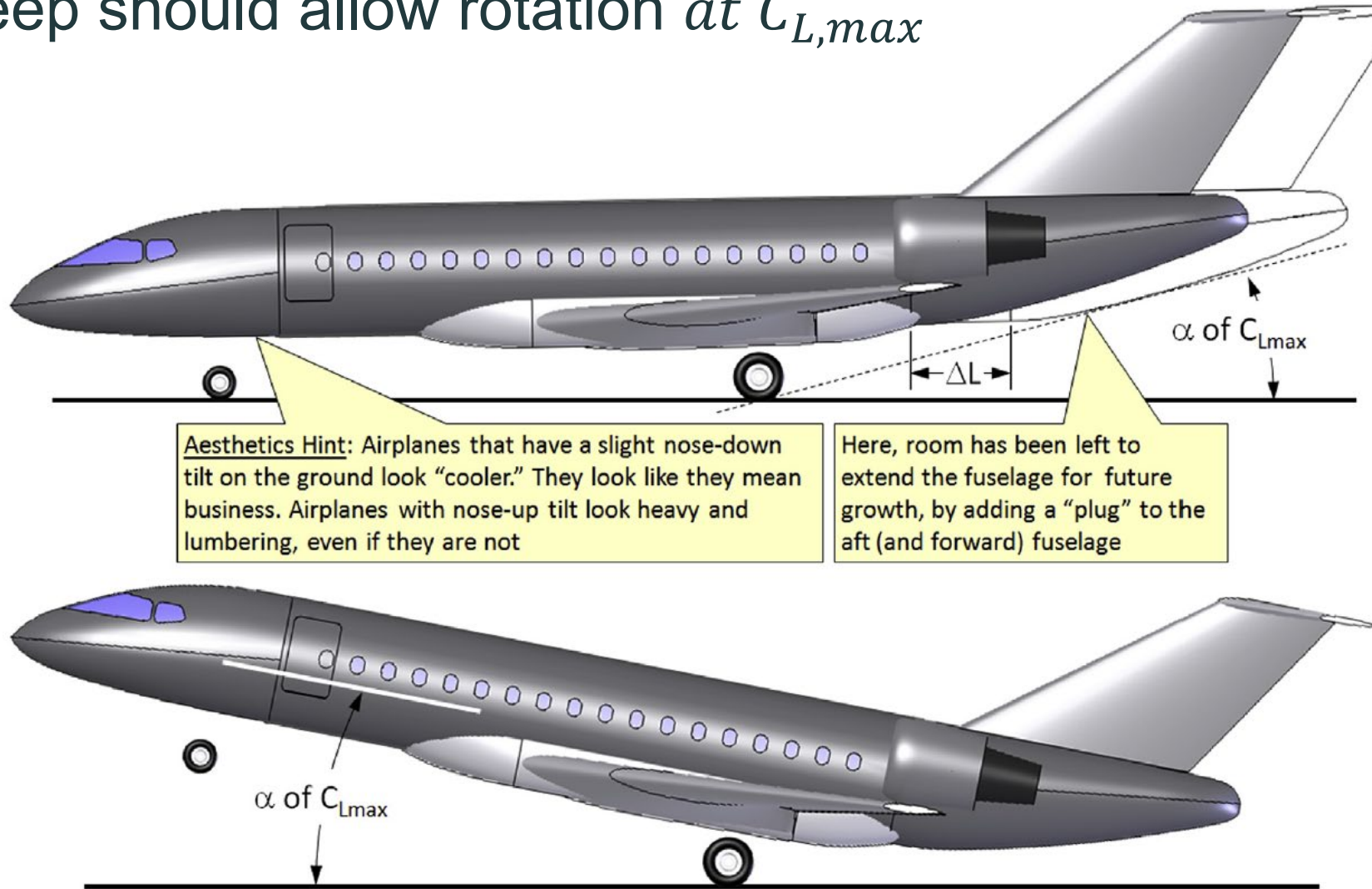
Definition of Geometric Fuselage Parameters

Currently Used Geometric Fuselage Parameters

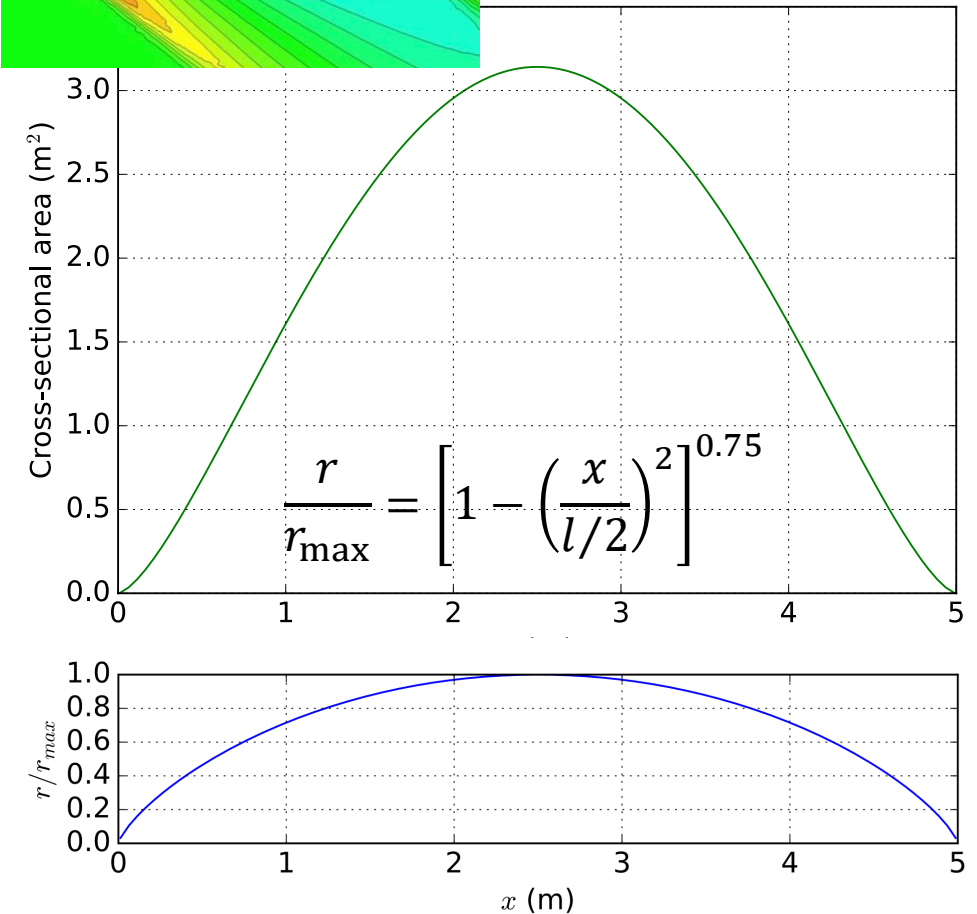
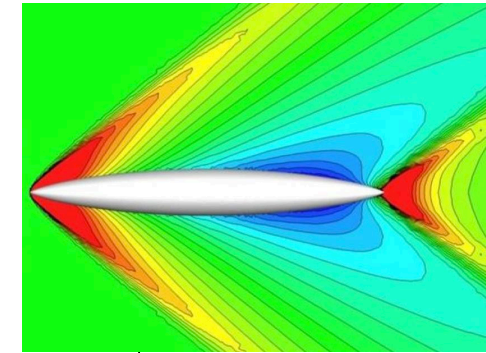
Airplane Type	l_f/d_f	l_{fc}/d_f	θ_{fc} (deg)
Homebuilts	4 - 8	3	2 - 9
Single Engine	5 - 8	3 - 4	3 - 9
Twins	3.6 - 8	2.6 - 4	6 - 13
Agricultural	5 - 8	3 - 4	1 - 7
Business Jets	7 - 9.5	2.5 - 5	6 - 11
Regionals	5.6 - 10	2 - 4	15 - 19
Jet Transports	6.8 - 11.5	2.6 - 4	11 - 16
Mil. Trainers	5.4 - 7.5	3	up to 14
Fighters	7 - 11	3 - 5	0 - 8
Mil. Transports, Bombers and Patrol Airplanes	6 - 13	2.5 - 6	7 - 25
Flying Boats	6 - 11	3 - 6	8 - 14
Supersonics	12 - 25	6 - 8	2 - 9

From Roskam, Airplane Design: Part 2 - preliminary configuration design and integration of the propulsion system

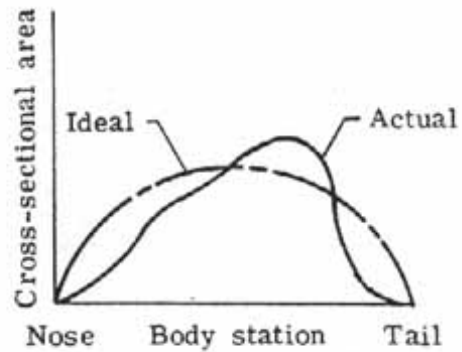
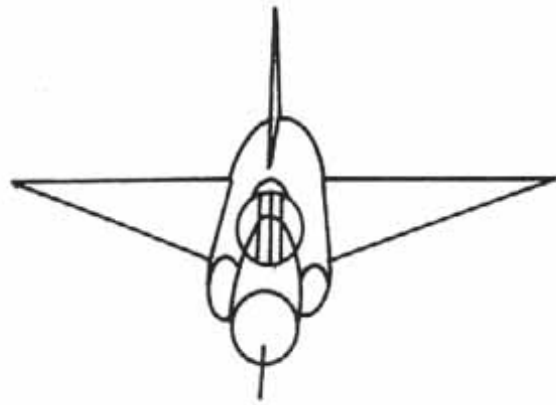
The upsweep should allow rotation *at $C_{L,max}$*



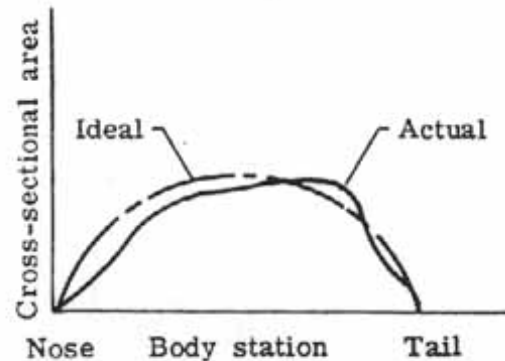
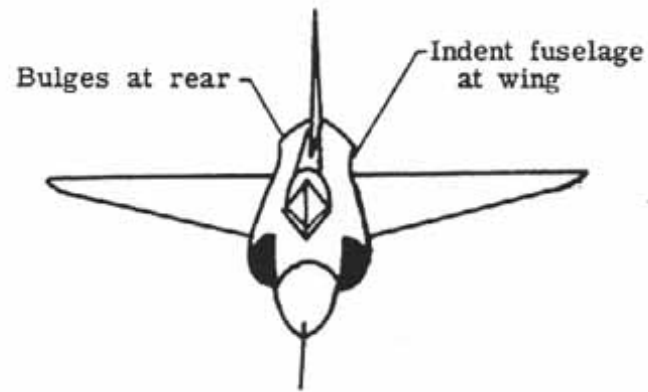
- ✈ Wave drag: caused by pressure drag due to shock waves.
- ✈ Wave drag increases with curvature, 2nd derivate of volume plot.
- ✈ The Sears-Hack body gives the lowest wave drag for $M=1$.
- ✈ Volume distribution in aircraft cannot follow that shape => where do we put the wings!



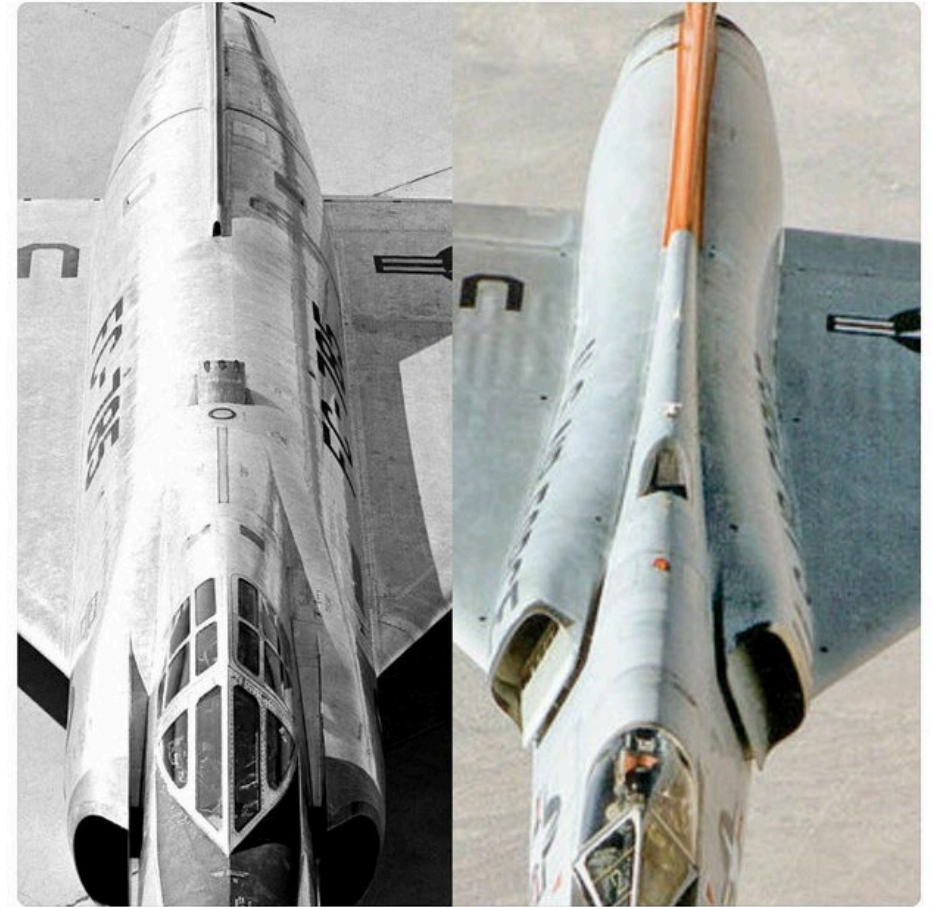
Supersonic area rule, reductions in wave drag up to 50%



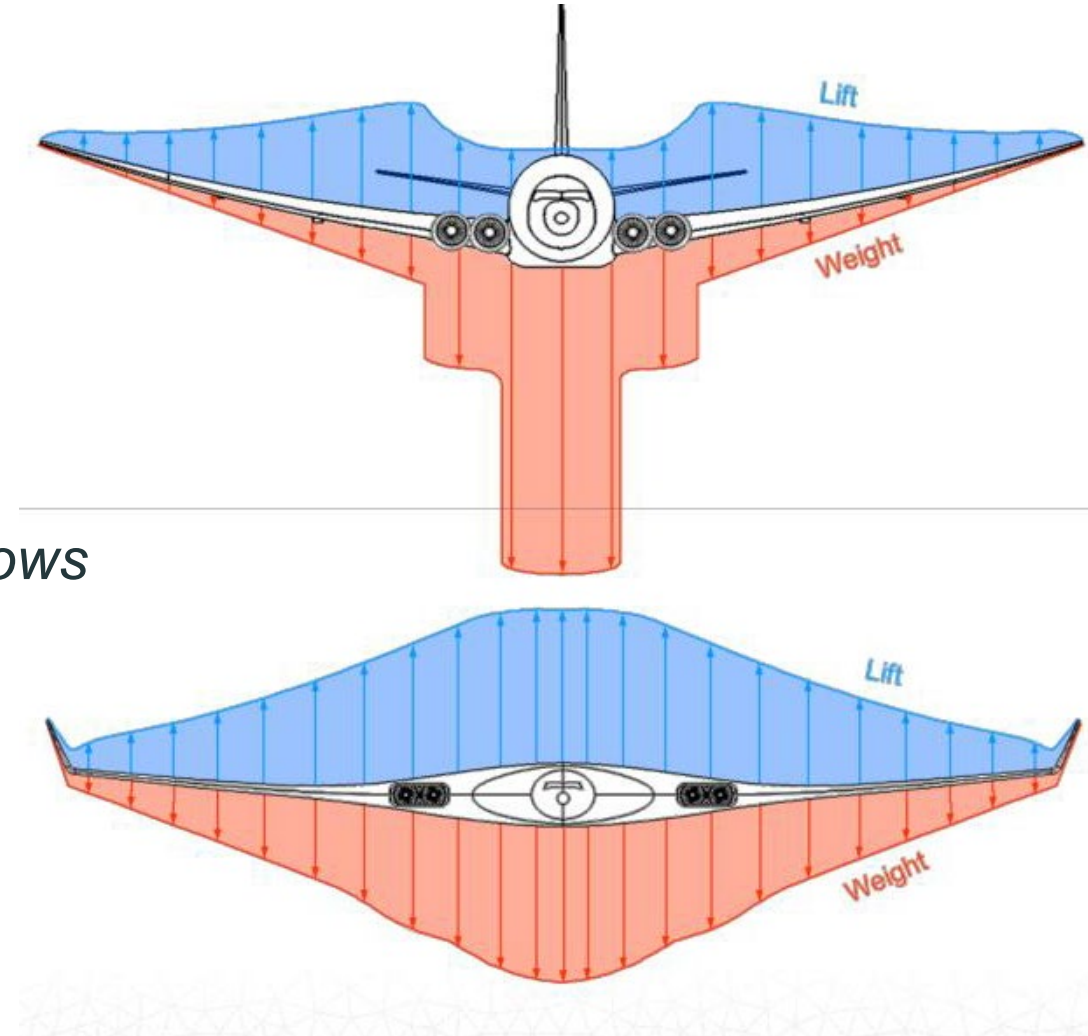
(a) YF-102A before area ruling.

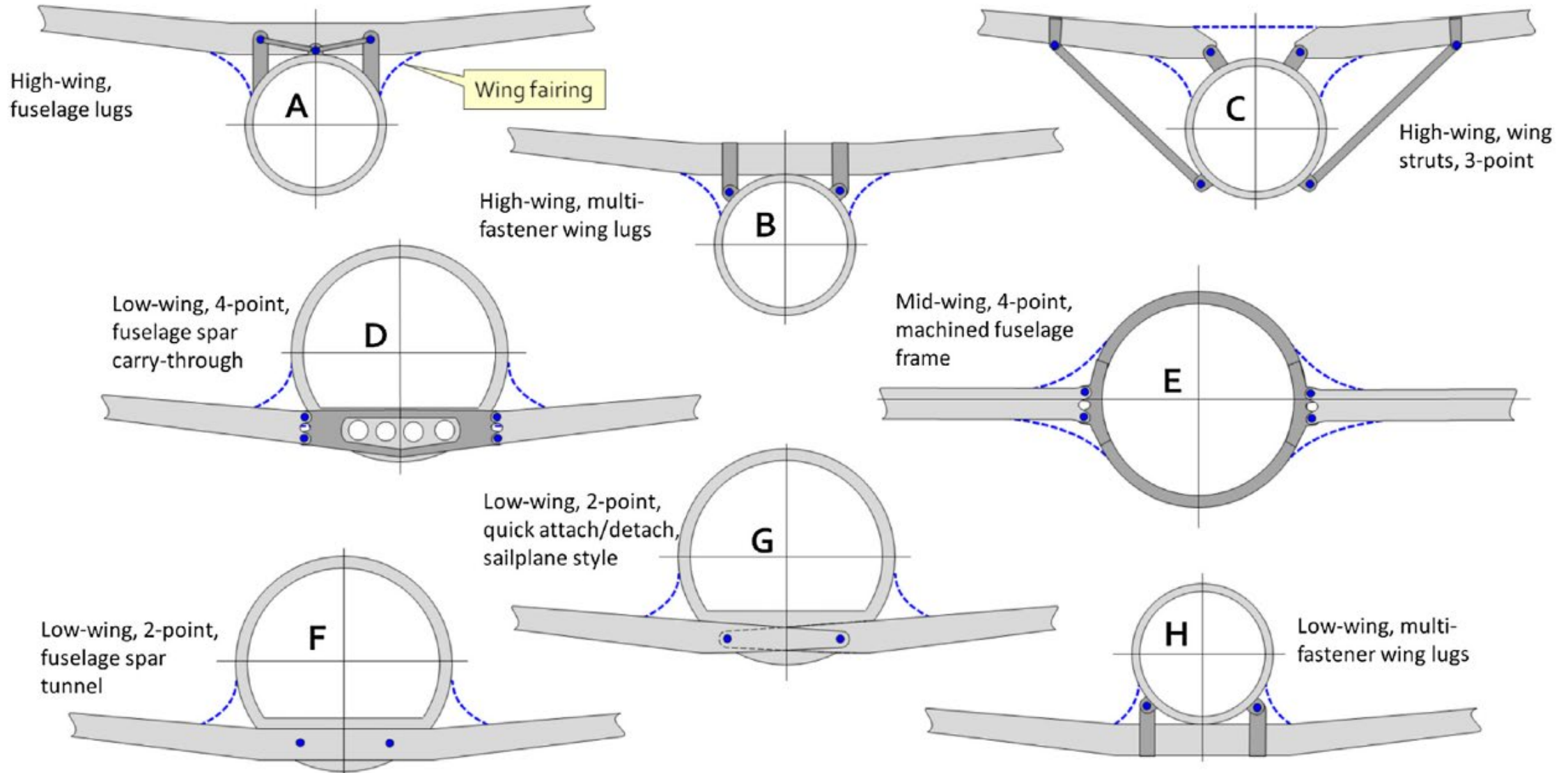


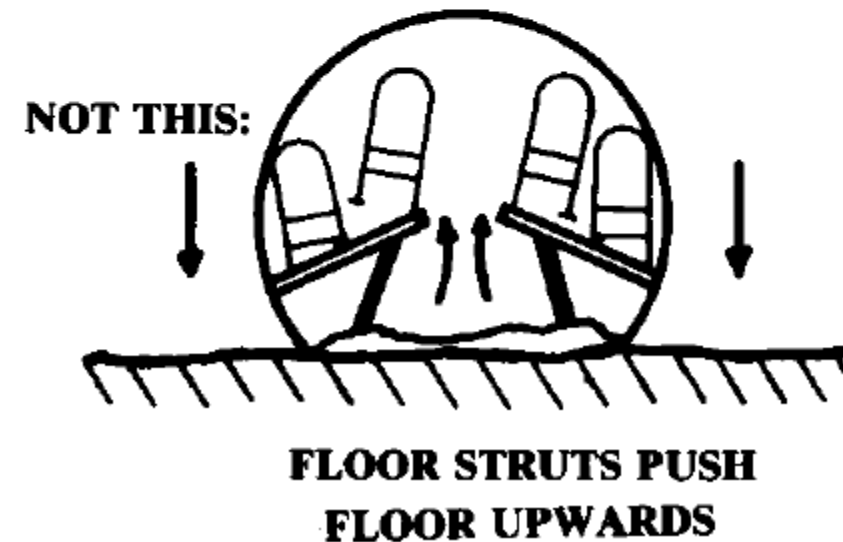
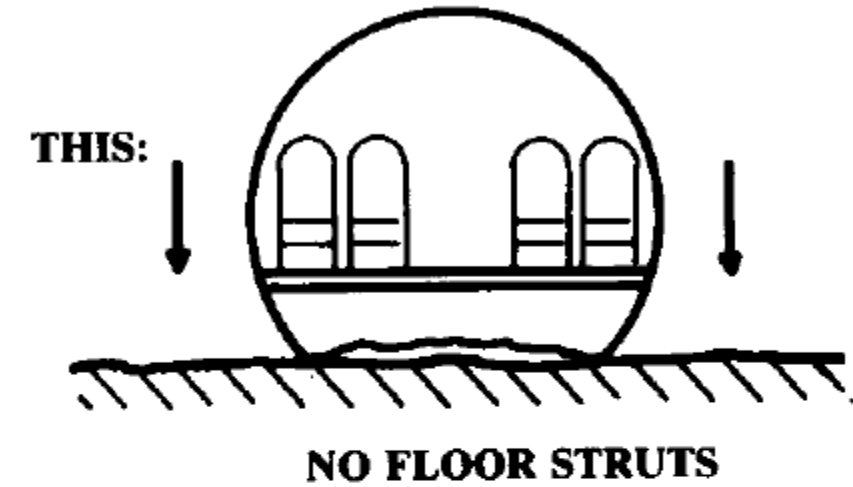
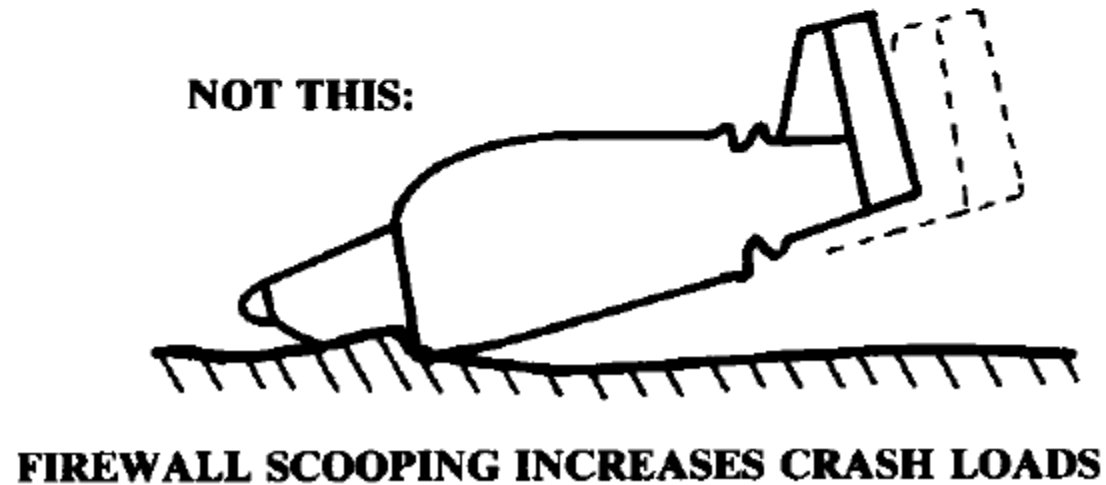
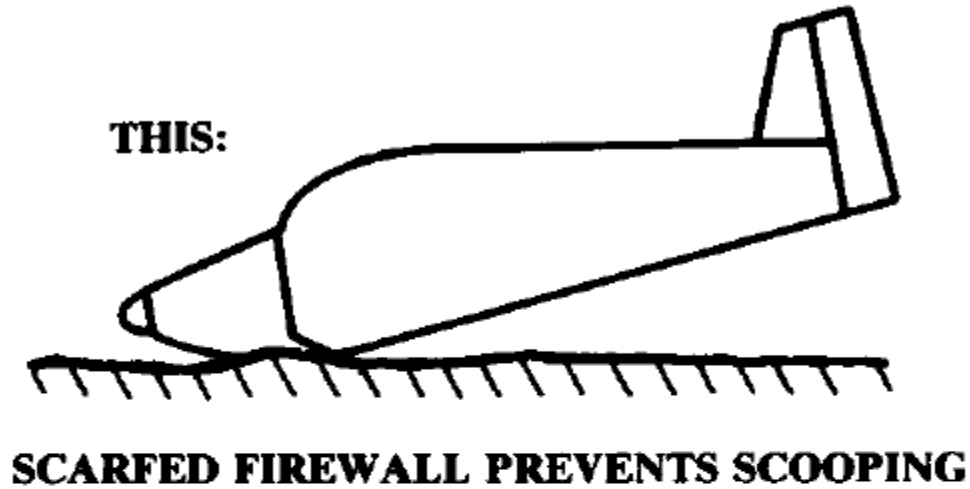
(b) F-102A after area ruling.



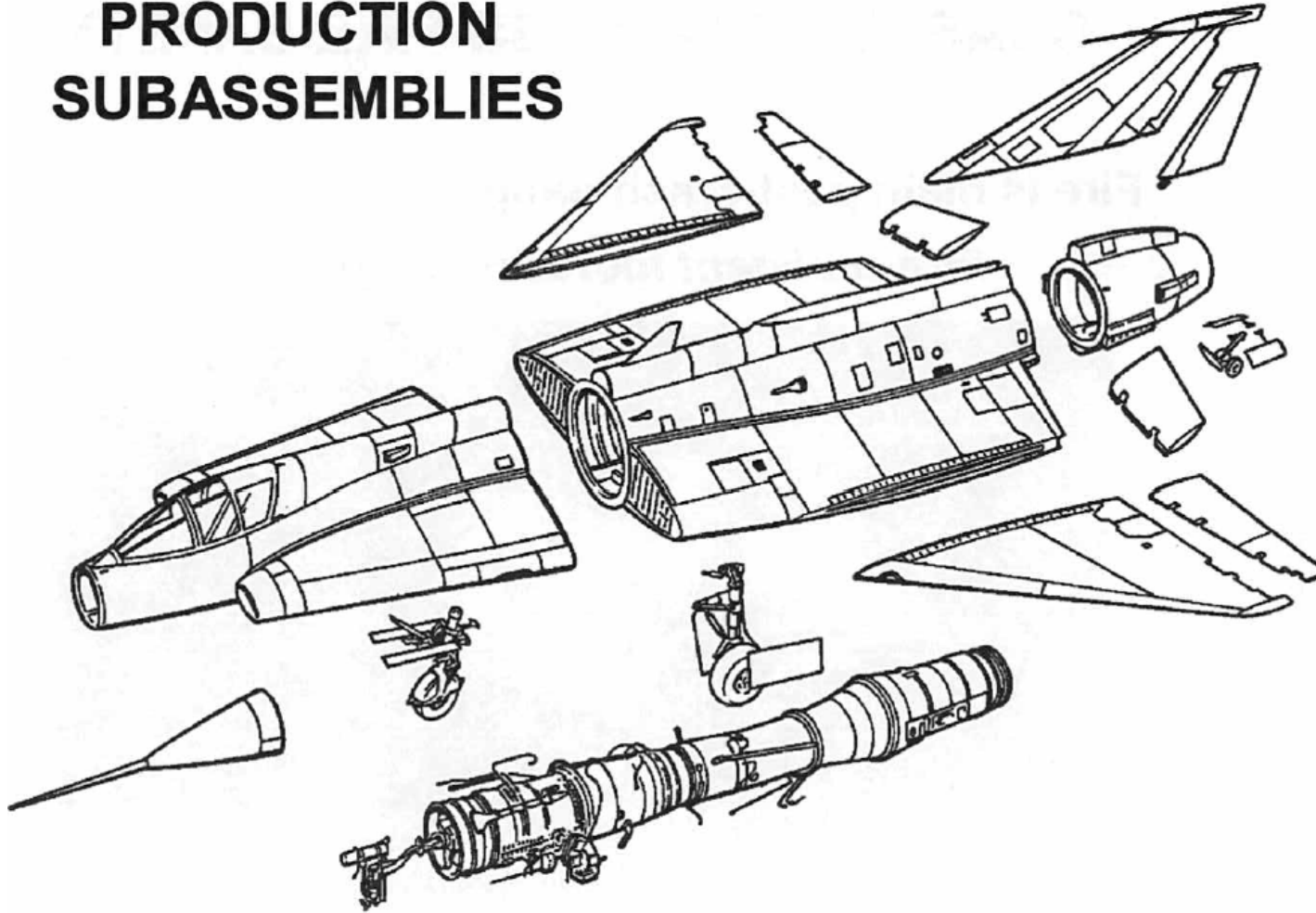
- ✈ Short, straight and continuous load paths.
 - Major loads on few major components
 - Heavy items near CG
- ✈ Minimum cutouts and doors.
 - Avoid engines buried in fuselage
 - Don't place the landing gear in the wingbox!
 - Beware of stress concentrations. *Why are windows rounded?*
- ✈ Avoid T-tails, H-tails and all moving tails.
- ✈ Span loading for weight reduction.
 - Distribute your loads in a smart way!



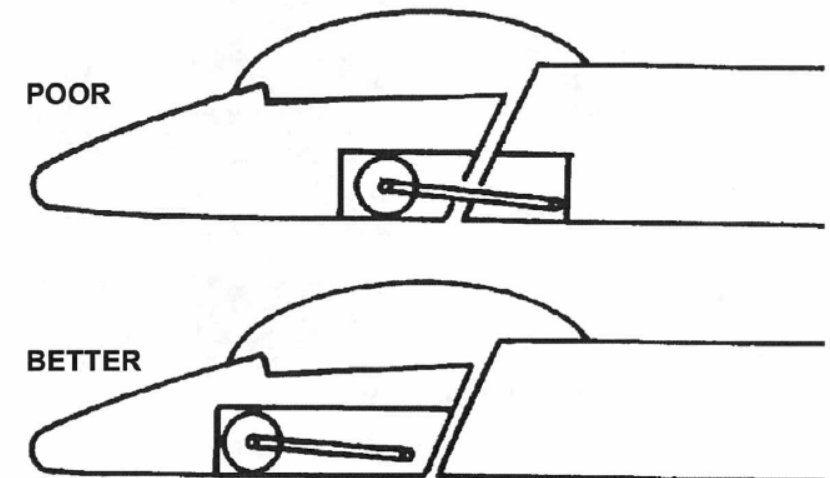




PRODUCTION SUBASSEMBLIES



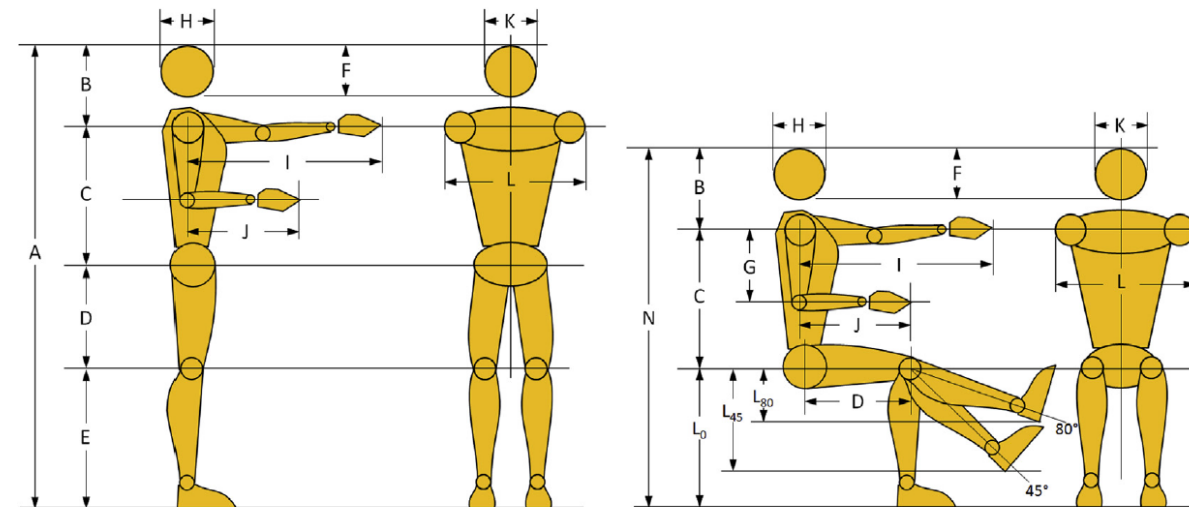
- ✈ Reduced part count.
- ✈ Accessibility
 - Mechanics need to stick their head in tight places
- ✈ Lower cost of materials and processes



Crew station, cabin and payload

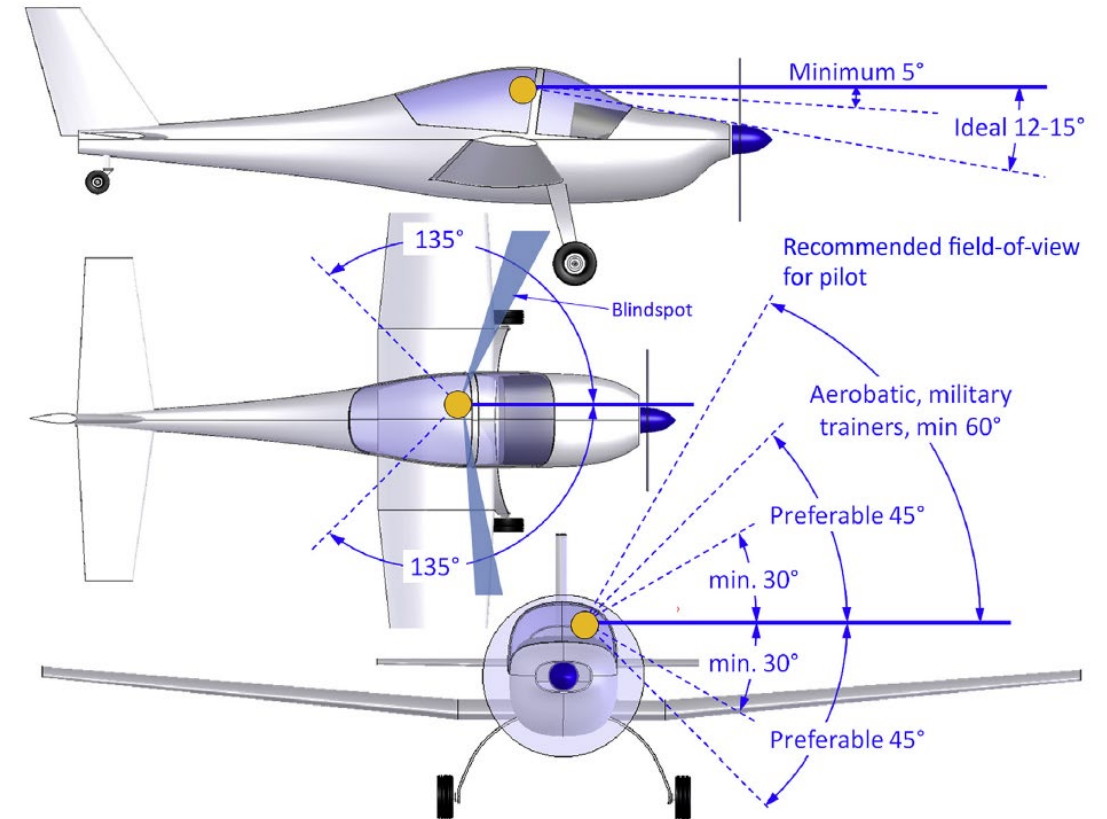
Important dimensions!

			MALE AND FEMALE PHYSICAL CHARACTERISTICS					
			1st	5th	50th	95th	99th	
Stature	A	Male	63.1	64.8	69.1	73.5	75.2	in
			160.3	164.6	175.5	186.7	191.0	cm
		Female	58.4	60.2	64.1	68.4	70.1	in
			148.3	152.9	162.8	173.7	178.1	cm
Shoulder-head	B	Male	11.8	11.9	12.5	13.0	13.2	in
			30.0	30.2	31.8	33.0	33.5	cm
		Female	10.9	11.3	11.8	12.3	12.5	in
			27.7	28.7	30.0	31.2	31.8	cm
Torso	C	Male	19	19.3	20.1	20.8	21.1	in
			48.3	49.0	51.1	52.8	53.6	cm
		Female	17.5	17.8	18.4	19.2	19.2	in
			44.5	45.2	46.7	48.8	48.8	cm
Hip height	D	Male	14.9	15.4	16.7	18.0	18.5	in
			37.8	39.1	42.4	45.7	47.0	cm
		Female	14.3	14.7	15.9	17.1	17.8	in
			36.3	37.3	40.4	43.4	45.2	cm
Knee height	E	Male	17.4	18.2	19.8	21.7	22.4	in
			44.2	46.2	50.3	55.1	56.9	cm
		Female	15.7	16.4	18.0	19.8	20.6	in
			39.9	41.7	45.7	50.3	52.3	cm
Head height	F	Male	8.4	8.6	8.6	9.1	9.4	in
			21.3	21.8	21.8	23.1	23.9	cm
		Female	7.8	8.3	8.6	9.1	9.4	in
			19.8	21.1	21.8	23.1	23.9	cm
Upper arm	G	Male	11.3	11.5	12.3	13.5	14	in
			28.7	29.2	31.2	34.3	35.6	cm
		Female	10.5	10.7	11.5	12.4	12.8	in
			26.7	27.2	29.2	31.5	32.5	cm
Head length	H	Male	7.1	7.3	7.8	8.2	8.4	in
			18.0	18.5	19.8	20.8	21.3	cm
		Female	6.8	7.0	7.4	7.8	8.0	in
			17.3	17.8	18.8	19.8	20.3	cm
Functional reach	I	Male	28.4	29.1	31.5	34.1	35.3	in
			72.1	73.9	80.0	86.6	89.7	cm
		Female	25.9	26.7	28.9	31.4	32.4	in
			65.8	67.8	73.4	79.8	82.3	cm
Elbow-fingertip	J	Male	17.1	17.6	19.2	20.6	21.3	in
			43.4	44.7	48.8	52.3	54.1	cm
		Female	15.4	16.0	17.4	19.0	19.6	in
			39.1	40.6	44.2	48.3	49.8	cm
Head width	K	Male	5.1	5.6	6.0	6.3	6.5	in
			13.0	14.2	15.2	16.0	16.5	cm
		Female	5.2	5.4	5.7	6.0	6.1	in
			13.2	13.7	14.5	15.2	15.5	cm
Shoulder breadth	L	Male	17.1	17.7	19.3	21.1	21.7	in
			43.4	45.0	49.0	53.6	55.1	cm
		Female	15	15.6	17.0	18.6	19.4	in
			38.1	39.6	43.2	47.2	49.3	cm

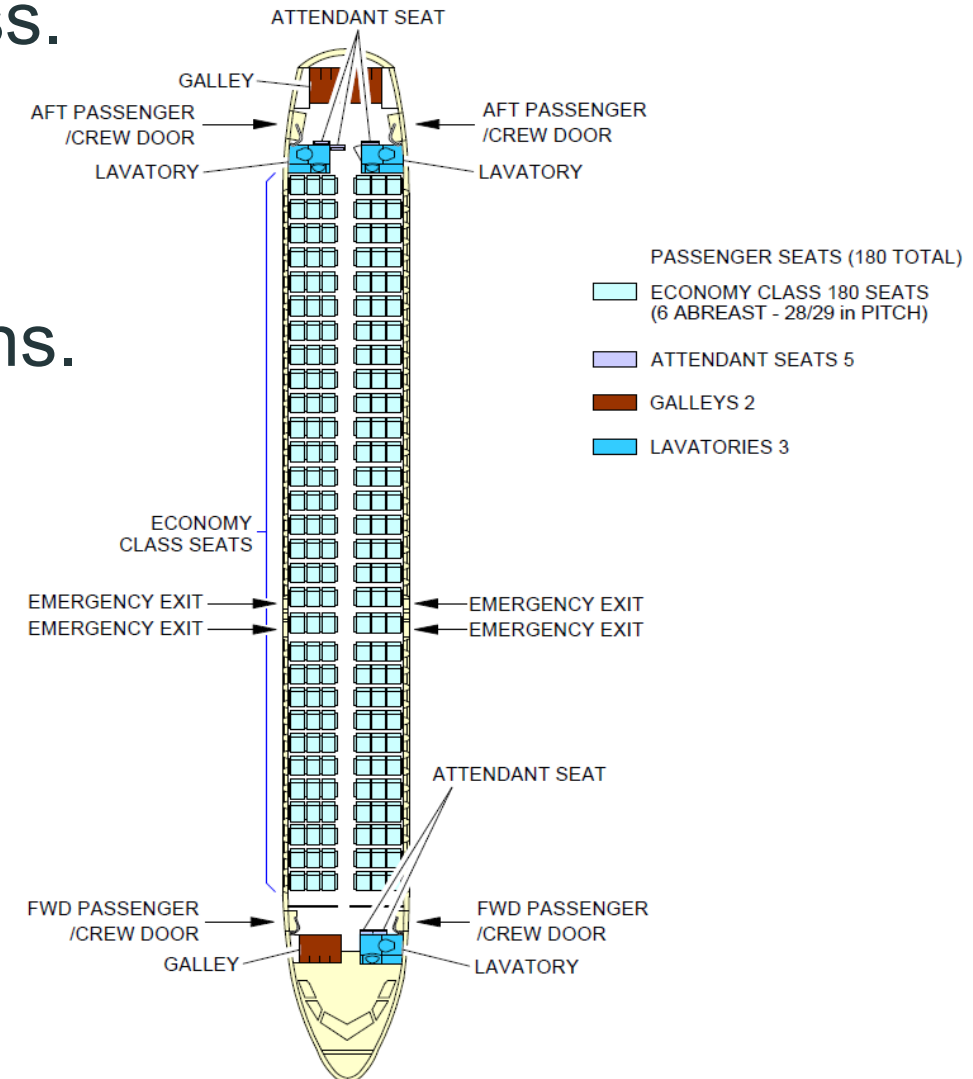


			MALE AND FEMALE PHYSICAL CHARACTERISTICS					
			1st	5th	50th	95th	99th	
Sitting stature	N	Male	48.2	49.4	52.4	55.5	56.7	in
			122.4	125.5	133.1	141.0	144.0	cm
		Female	44.1	45.5	48.2	51.3	52.3	in
			112.0	115.6	122.4	130.3	132.8	cm
Leg at 0°	L ₀	Male	17.4	18.2	19.8	21.7	22.4	in
			44.2	46.2	50.3	55.1	56.9	cm
		Female	15.7	16.4	18	19.8	20.6	in
			39.9	41.7	45.7	50.3	52.3	cm
Leg at 45°	L ₄₅	Male	12.3	12.9	14.0	15.3	15.8	in
			31.3	32.7	35.6	39.0	40.2	cm
		Female	11.1	11.6	12.7	14.0	14.6	in
			28.2	29.5	32.3	35.6	37.0	cm
Leg at 80°	L ₈₀	Male	3.0	3.2	3.4	3.8	3.9	in
			7.7	8.0	8.7	9.6	9.9	cm
		Female	2.7	2.8	3.1	3.4	3.6	in
			6.9	7.2	7.9	8.7	9.1	cm

- ✈ Defines the fuselage shape in the vicinity of the cockpit.
- ✈ Mainly regulated by vision requirements
- ✈ For sizing look into existing aircraft.

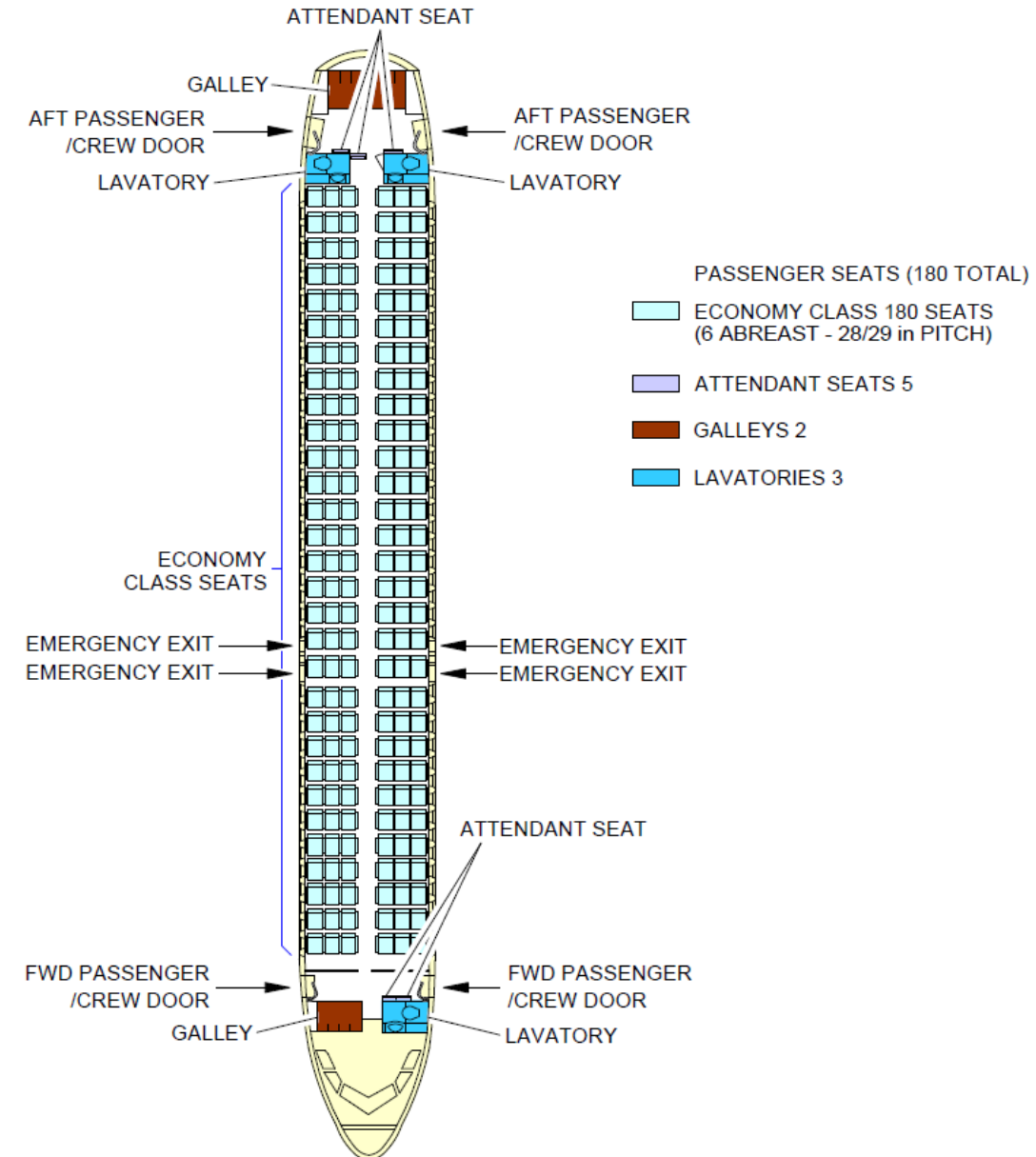
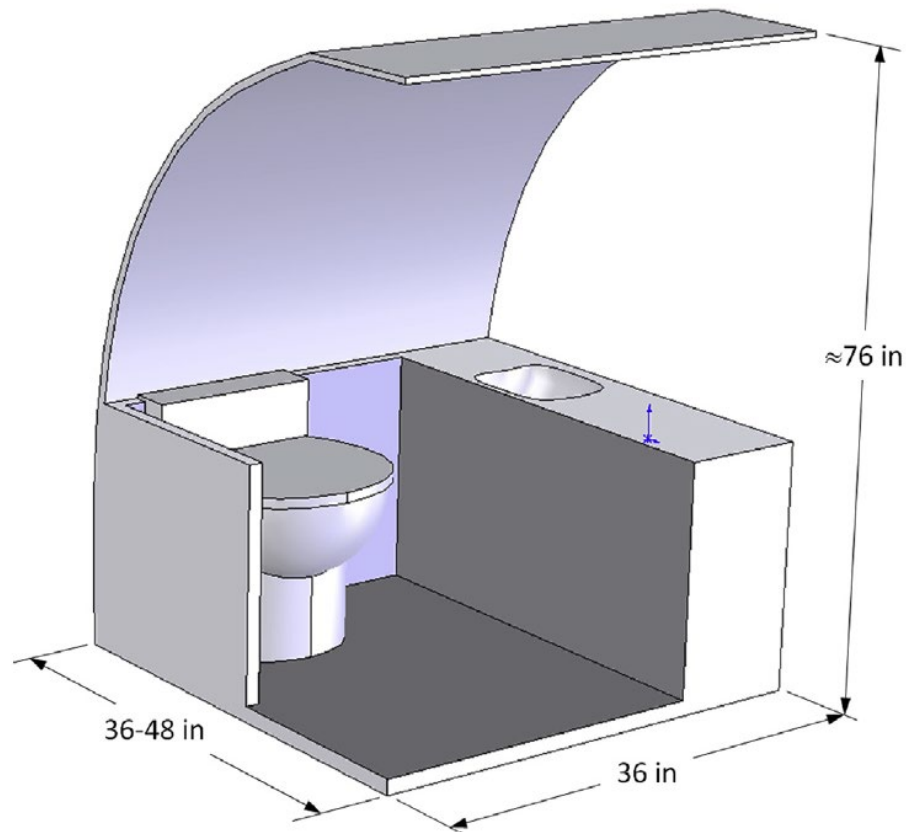


- ✈ Number of classes and allowances per class.
- ✈ Space for galley and toilets.
- ✈ Escape provisions.
- ✈ Good idea to look into existing configurations.
- ✈ For long range you need space for crew quarters



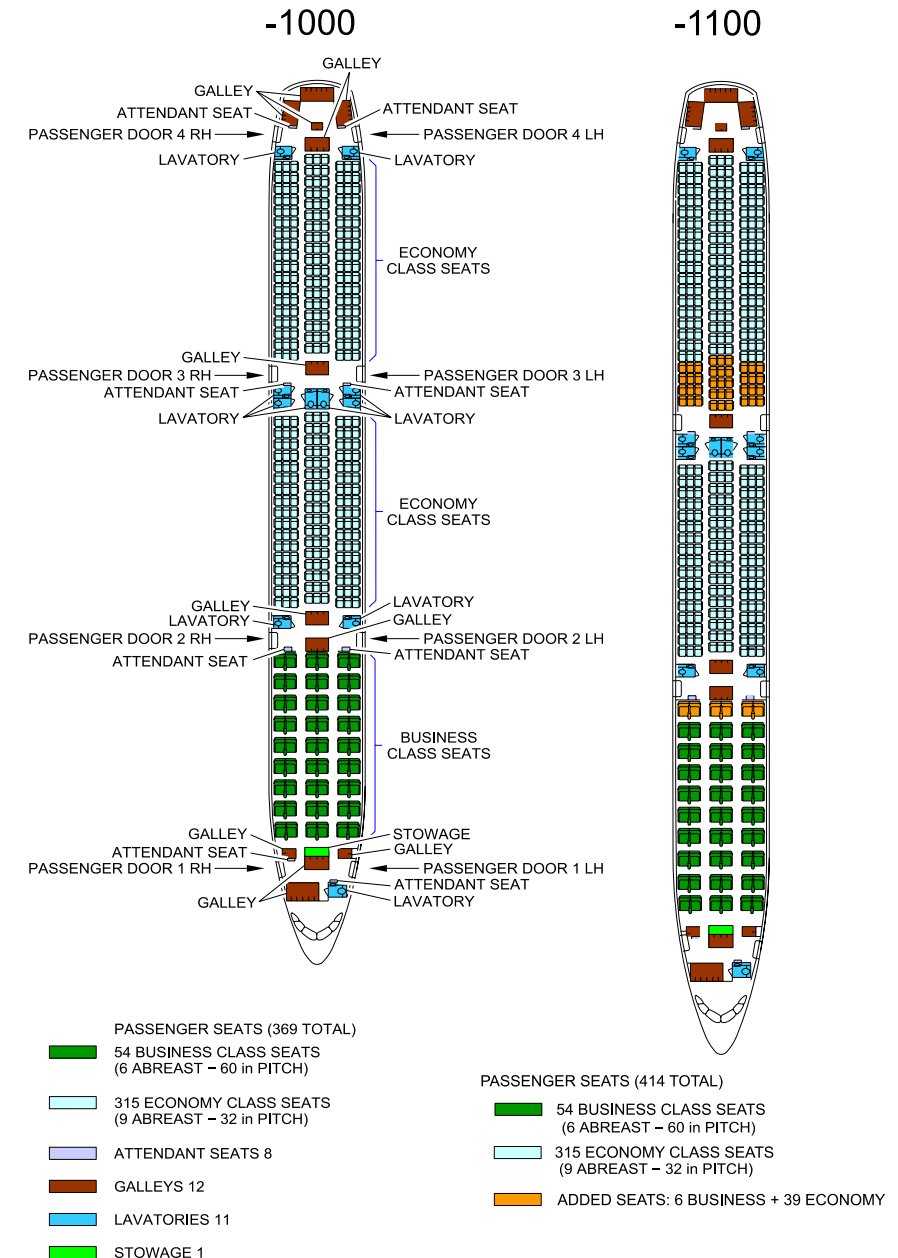
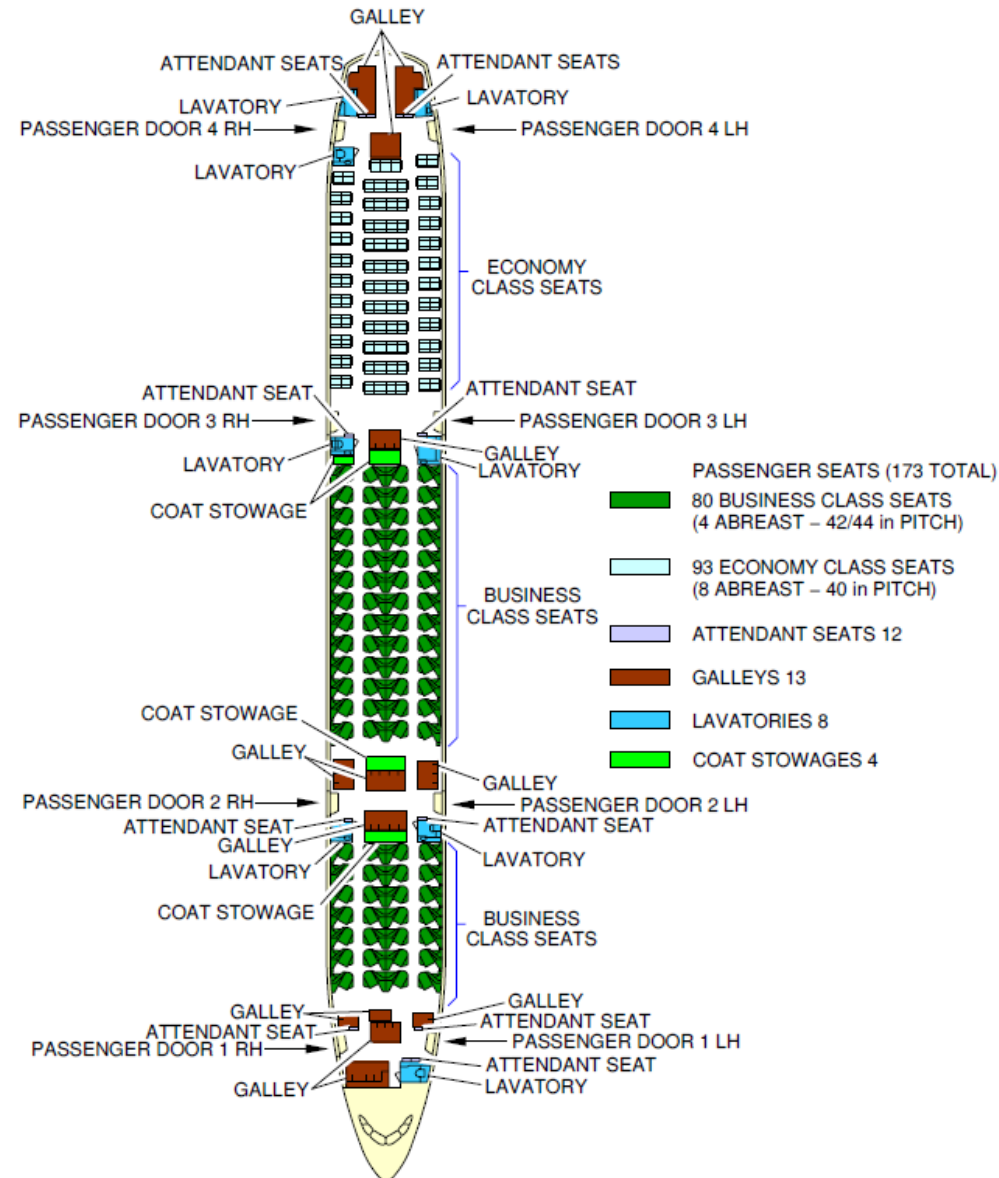
✈ Use manufacture, airline data.

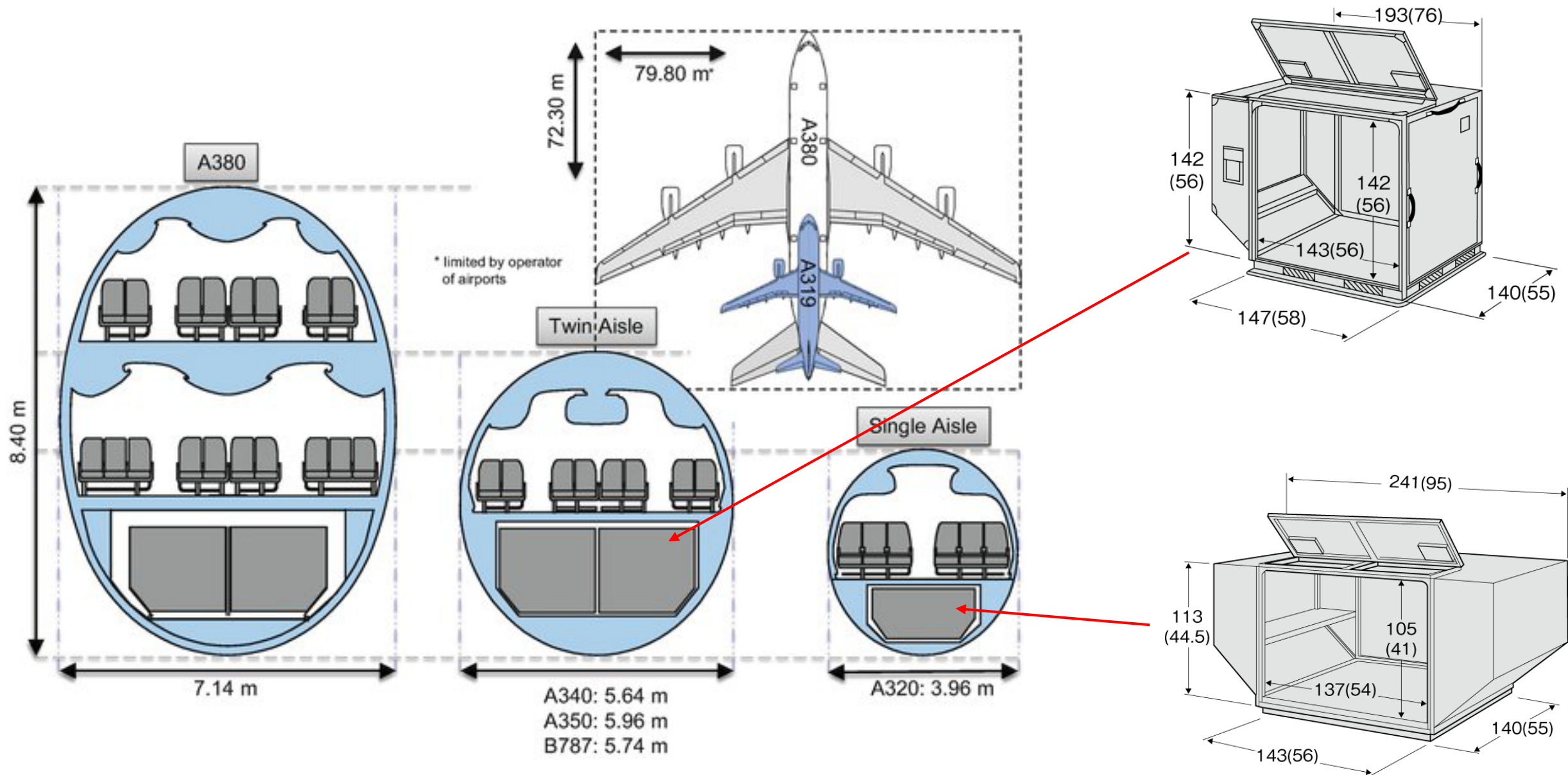
- https://www.boeing.com/commercial/airports/planning_manuals.page
- <https://www.airbus.com/en/airport-operations-and-technical-data/aircraft-characteristics>

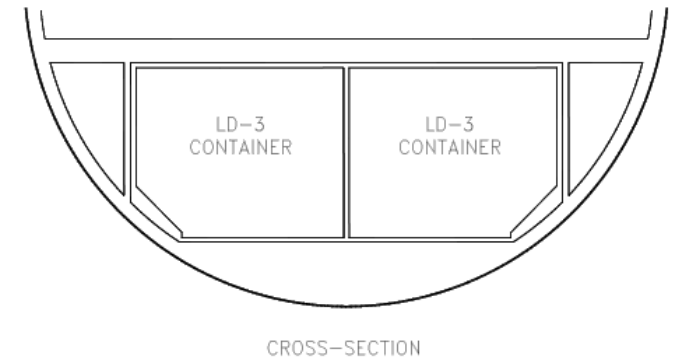
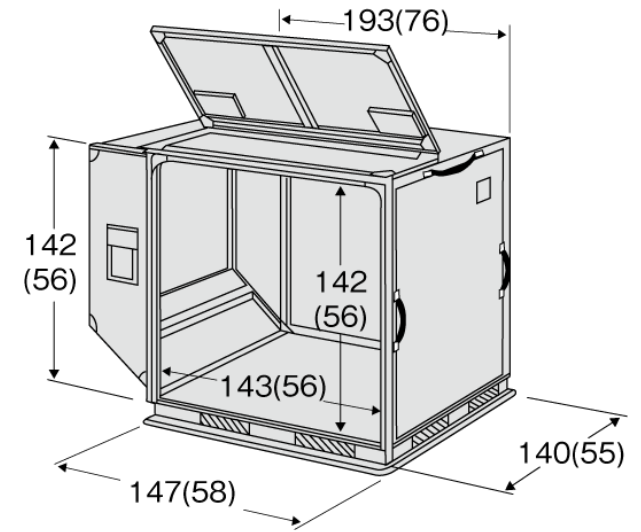
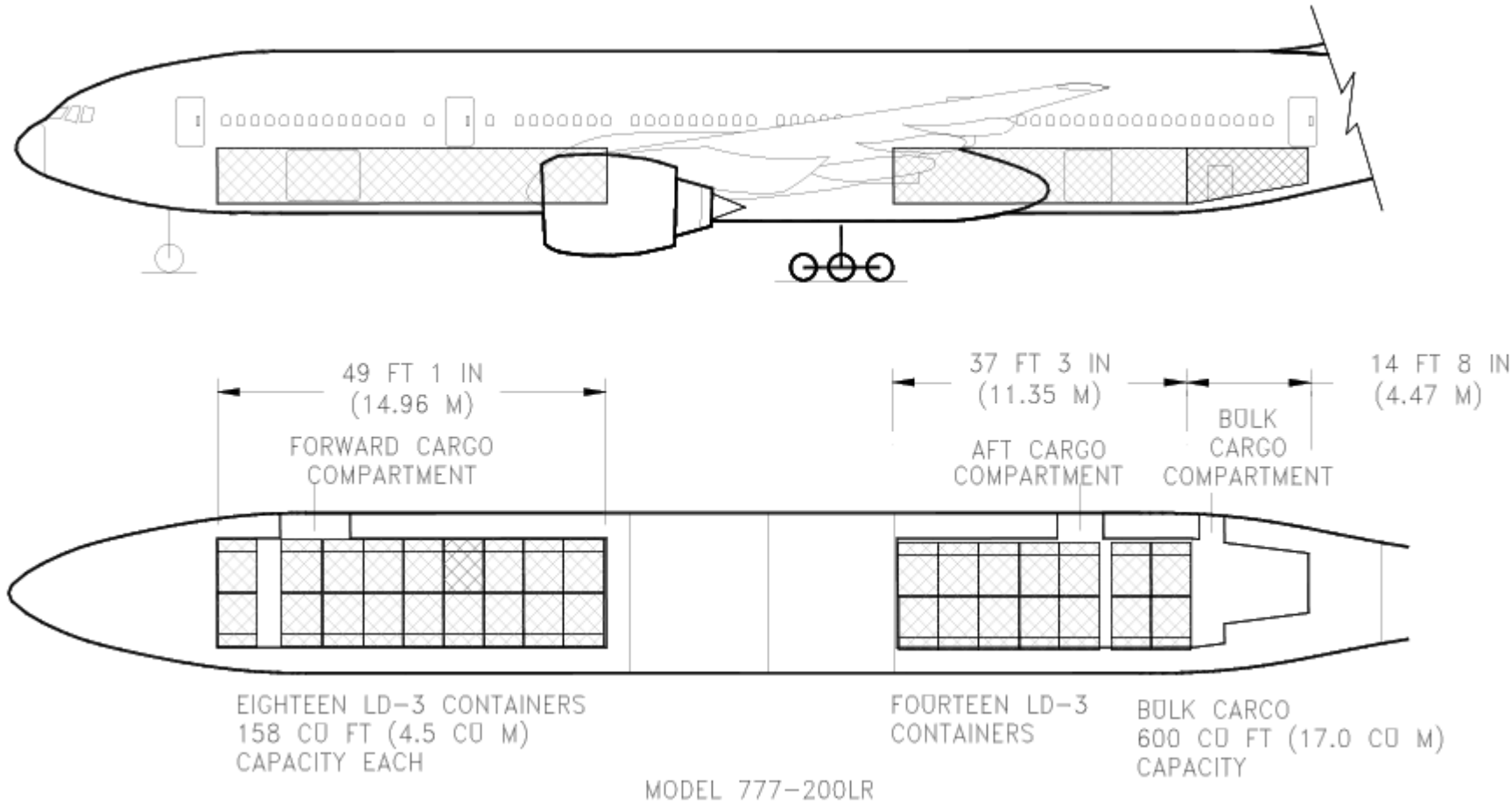


Passenger compartment – allowances LR

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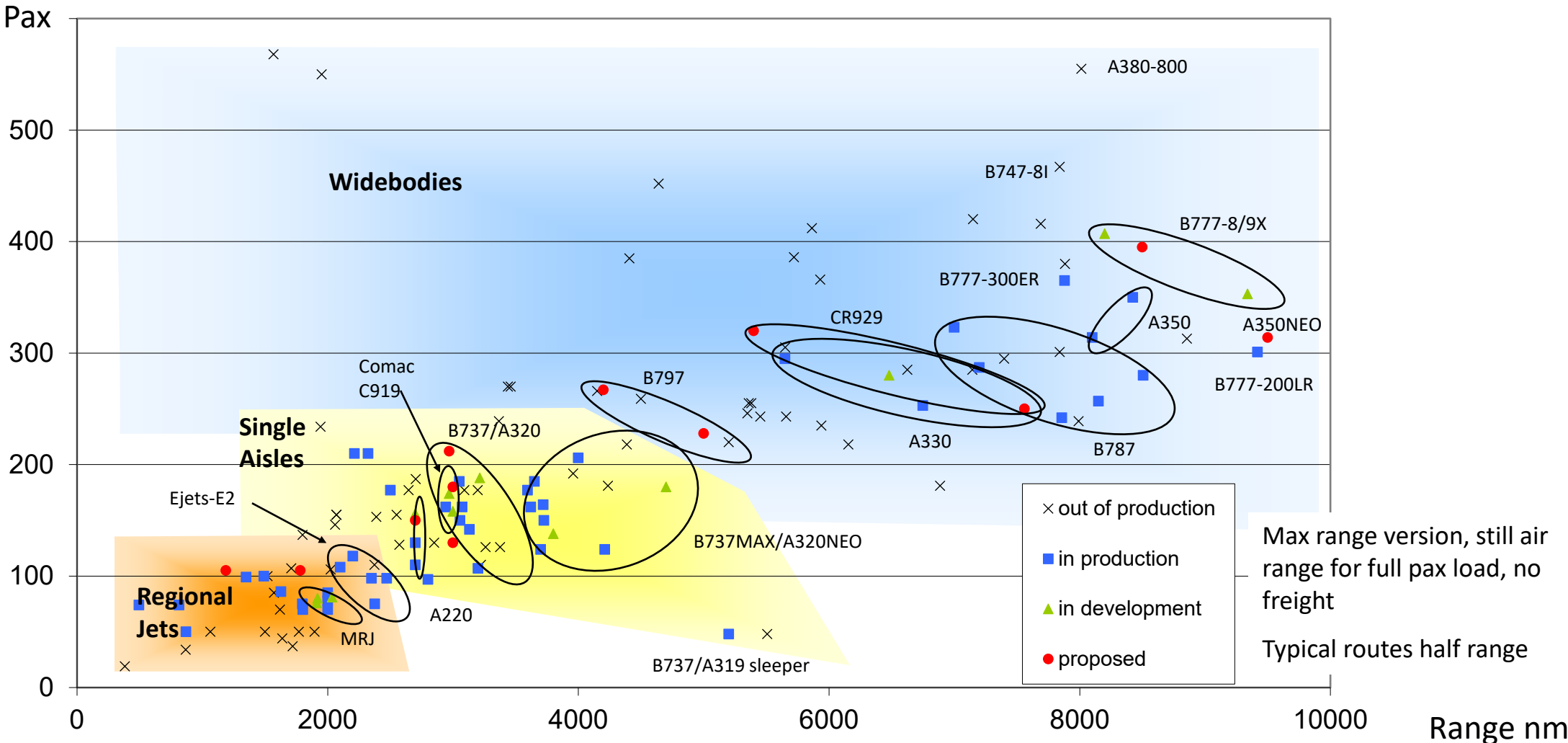






Capacity-range of aircraft

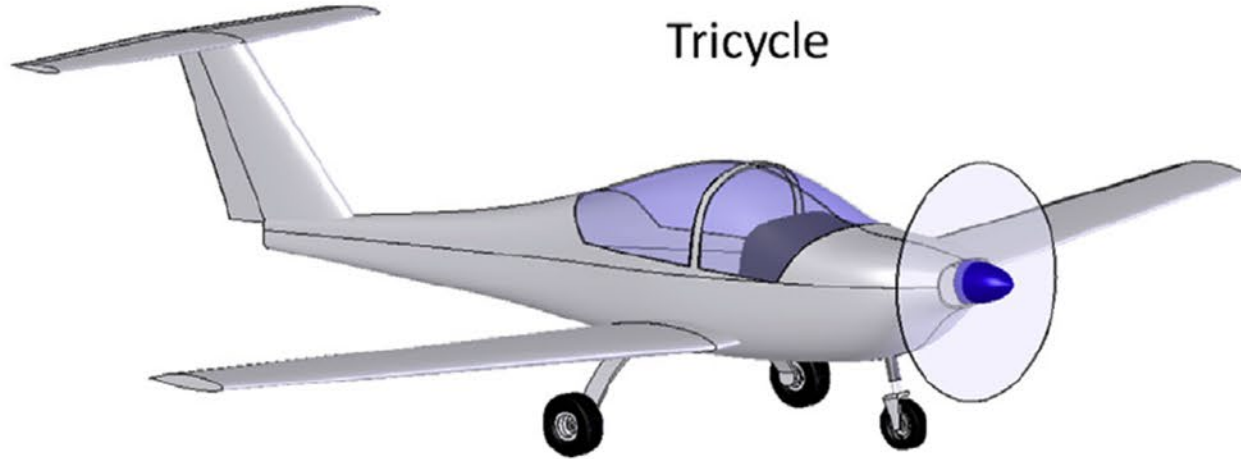
Manufacturer's typical seating



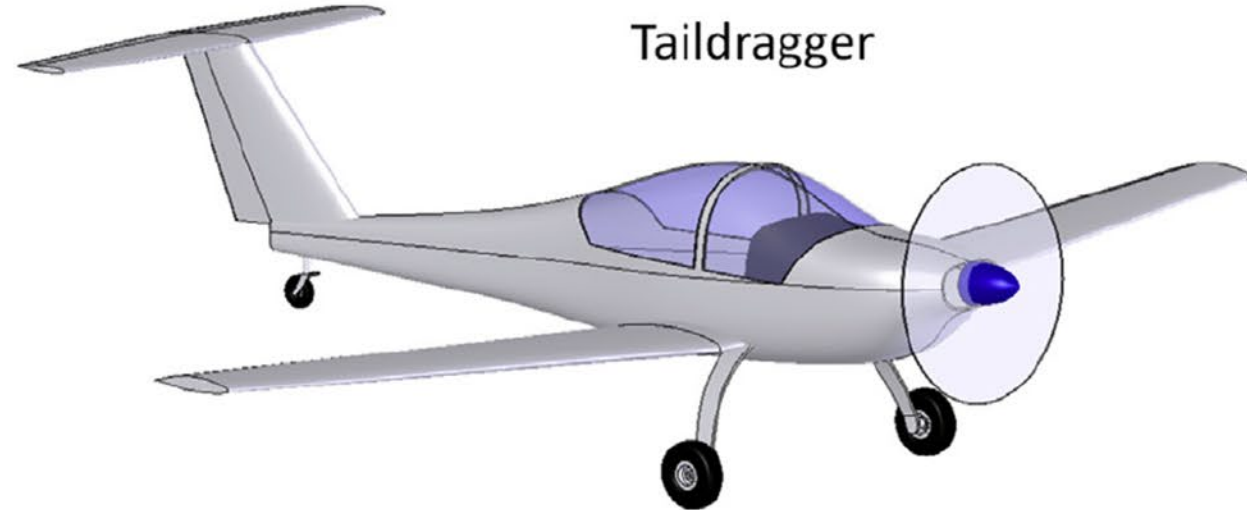
Landing gear



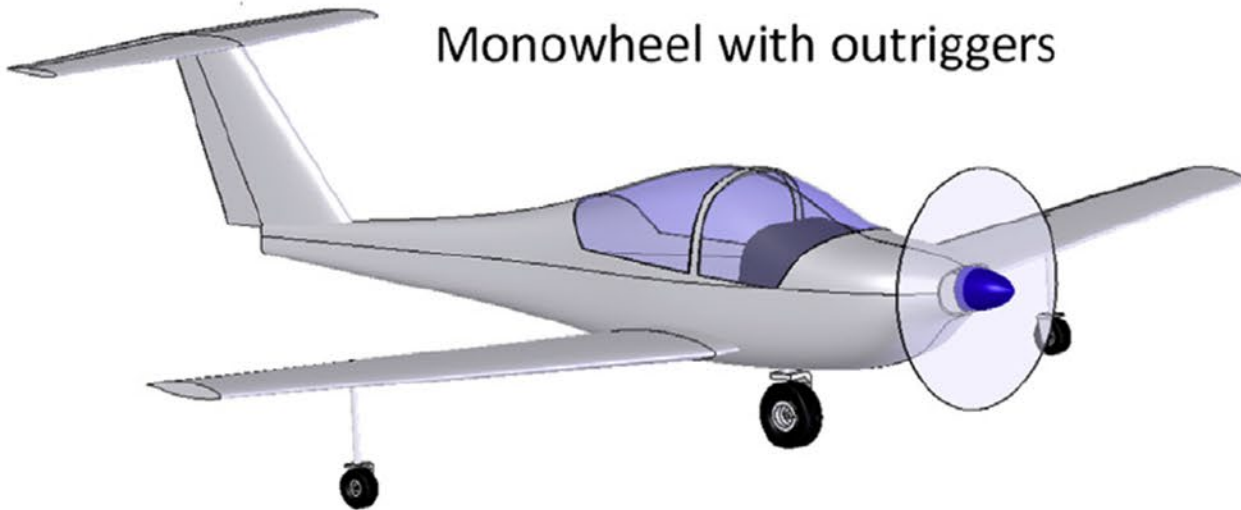
Tricycle



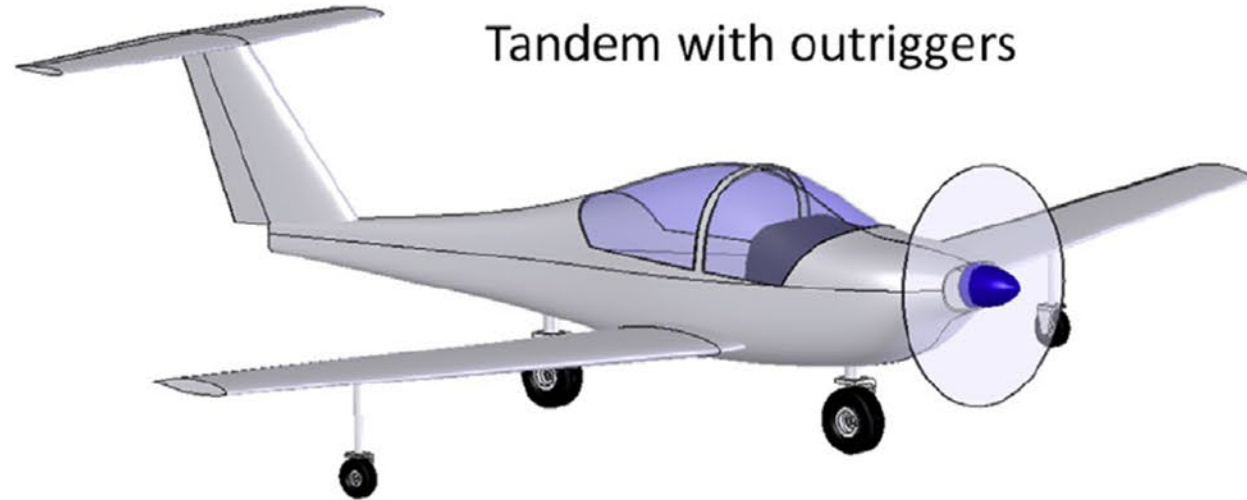
Taildragger



Monowheel with outriggers

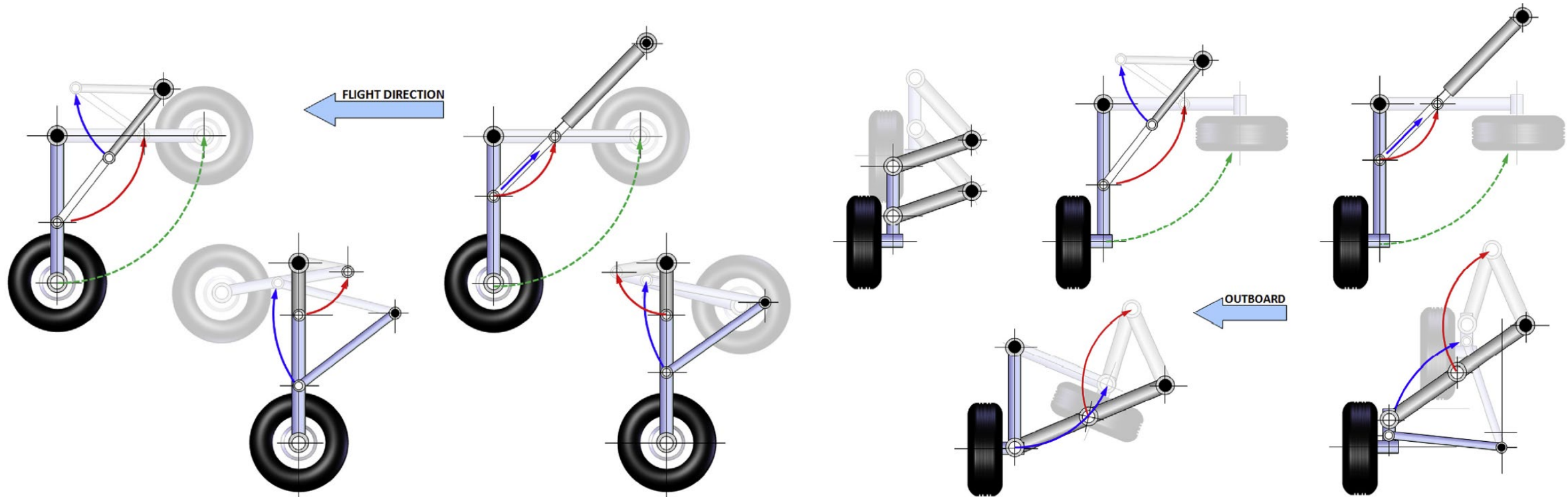


Tandem with outriggers



Landing gear – Retraction (make sure it fits!)

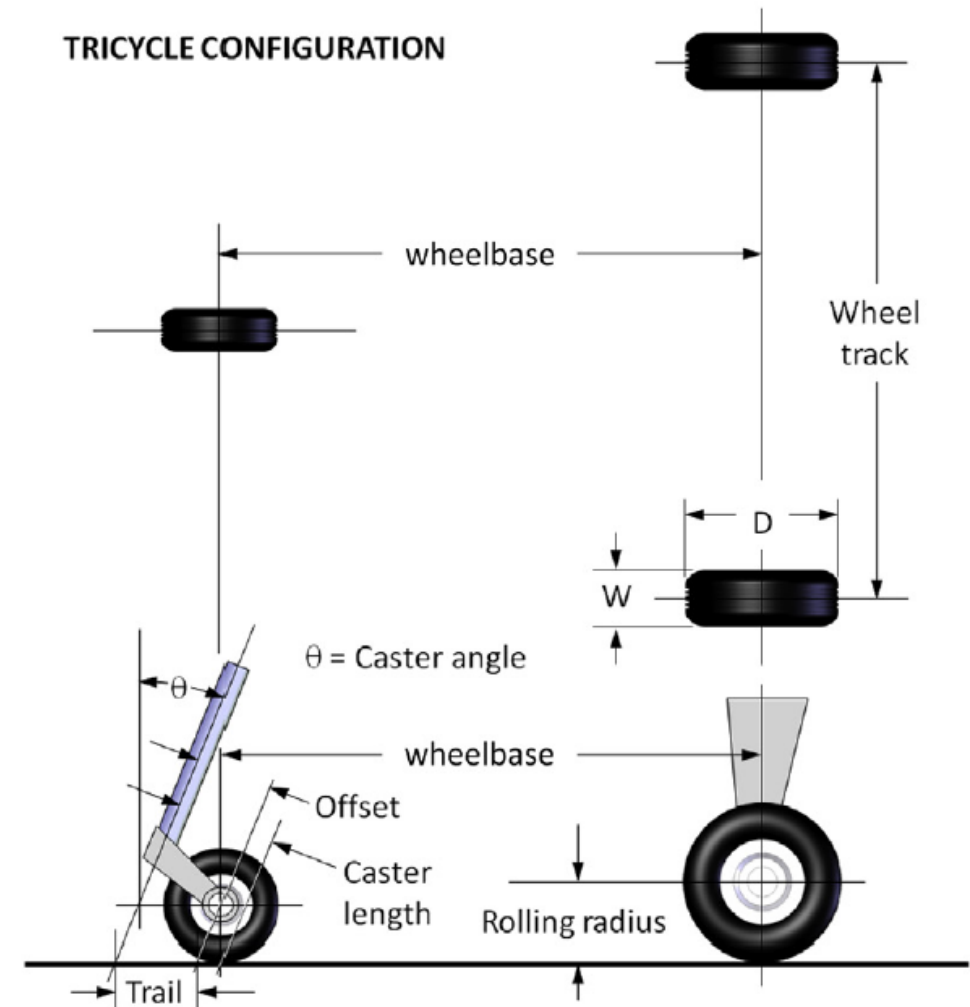
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- ✈ For sizing the main tires use statistical data (Tab 11.1).

$AW_{on\ wheel}^B$				
	Diameter		Width	
	A	B	A	B
General aviation	5.1	0.349	2.3	0.312
Buisness twin	8.3	0.251	3.5	0.216
Transport	5.3	0.315	0.39	0.480

- The main tires carry about 90% of the weight
 - Nose tires carry at least 10%, but no more than 20%.
- ✈ Nose wheels vary between 60% - 100% of main



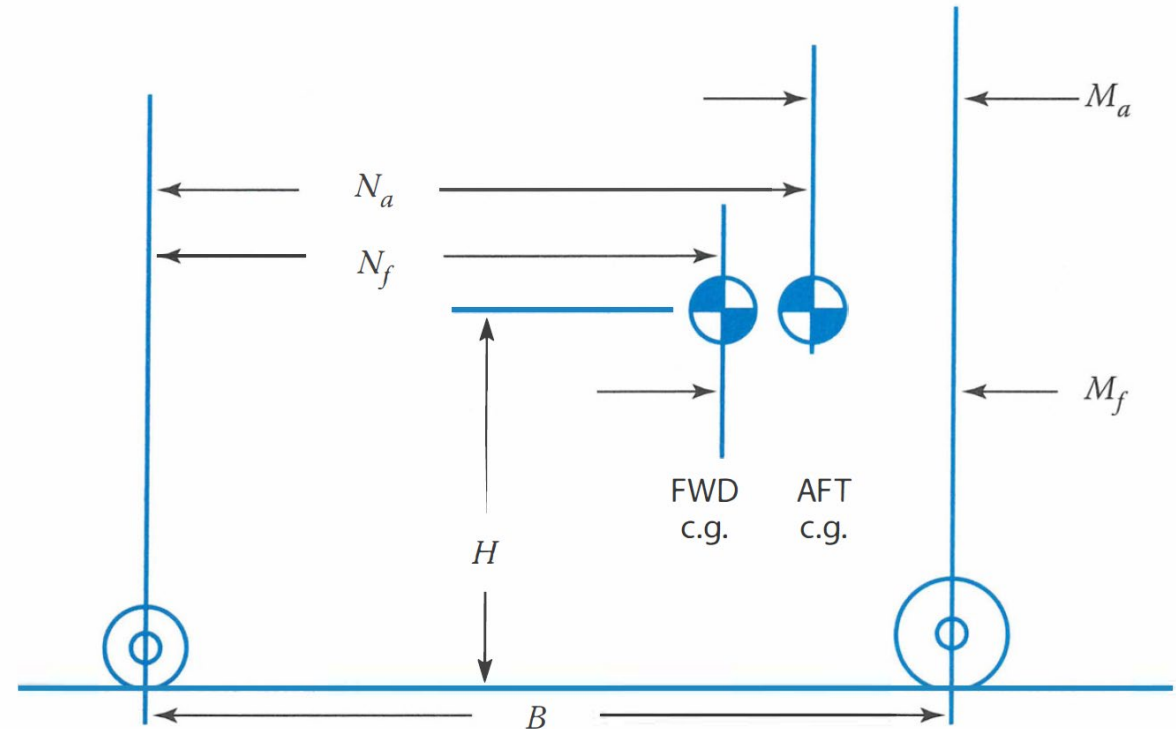
- ✈ In the final layout choose from a catalog (Tab. 11.2).
 - Requires a CG envelope
 - Make sure the size can withstand the static and dynamic loads.
 - FAR25 requires a 7% margin in the calculated loads

$$\text{Max Static load} = W_0 \frac{N_a}{B} g_0 \text{ [N]}$$

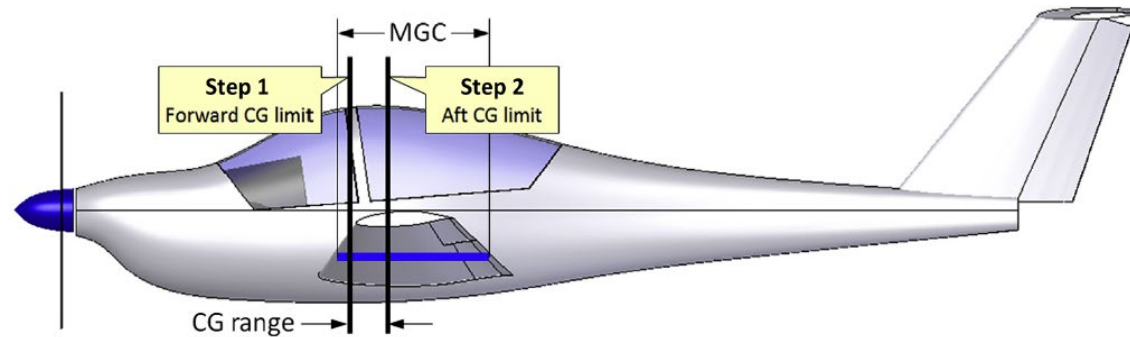
$$(\text{Max Static load})_{\text{nose}} = W_0 \frac{M_f}{B} g_0 \text{ [N]}$$

$$(\text{Min Static load})_{\text{nose}} = W_0 \frac{M_a}{B} g_0 \text{ [N]}$$

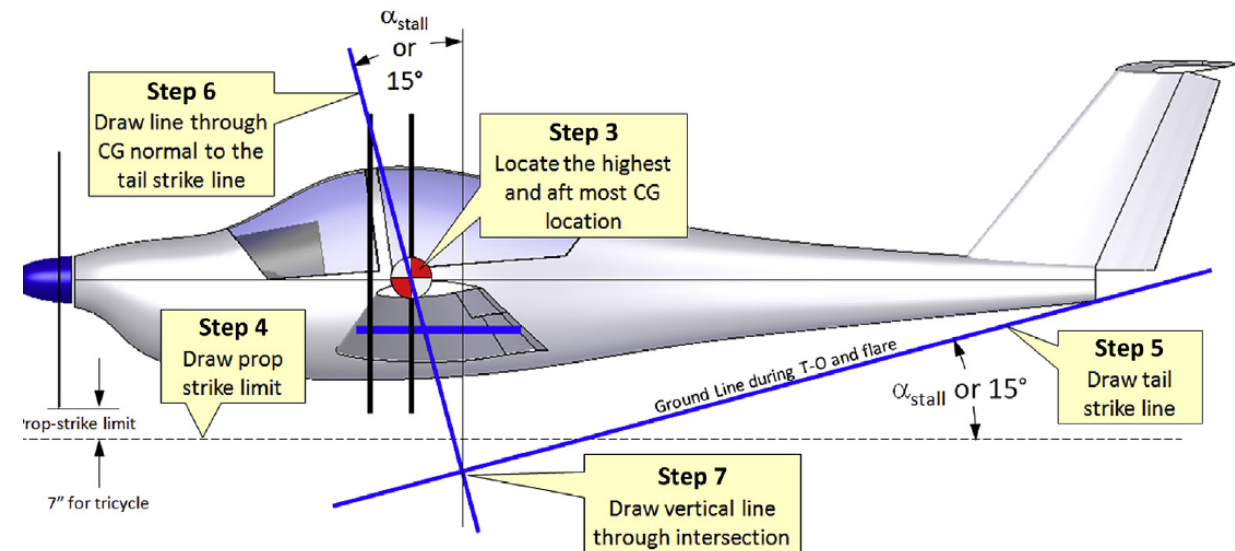
$$(\text{Dynamic brake load})_{\text{nose}} = \frac{3HW_0}{B} \text{ [N]}$$



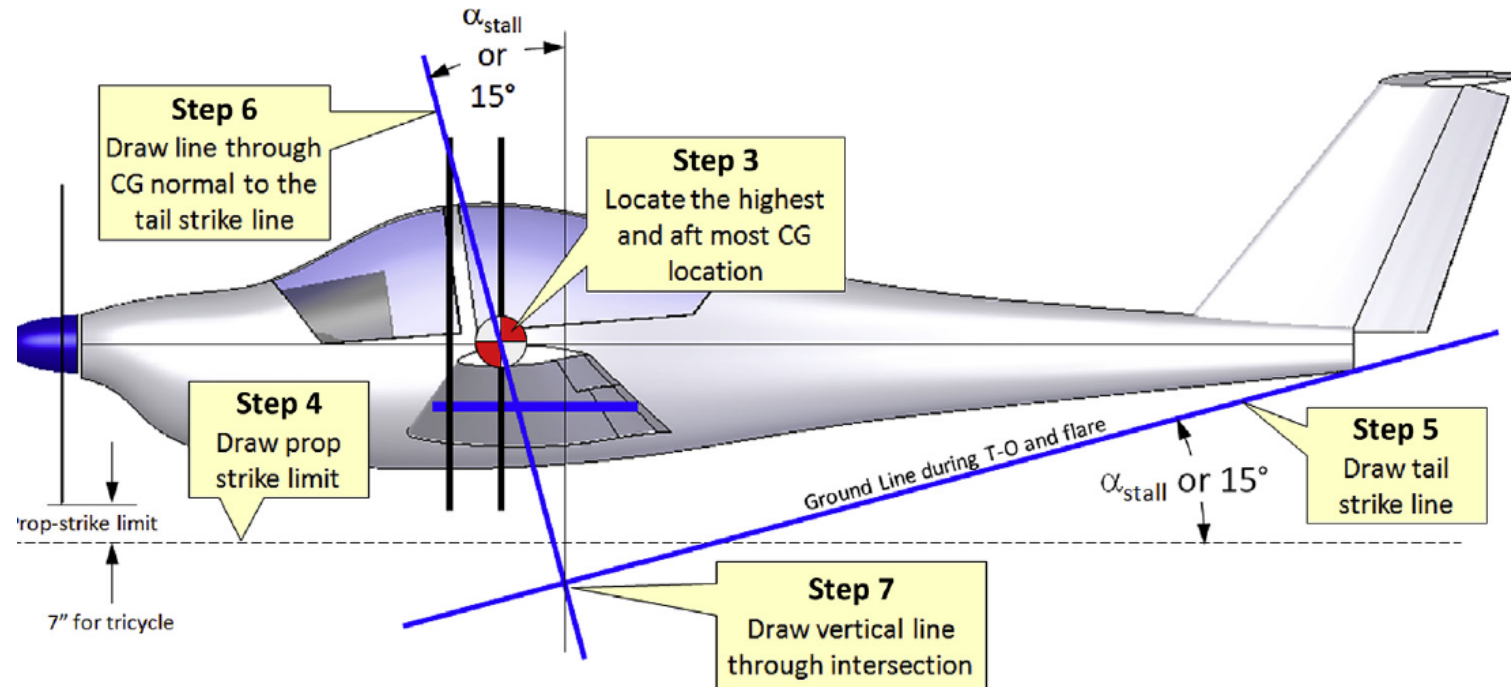
- ✈ Step 1 & Step 2: Draw forward and aft CG limit



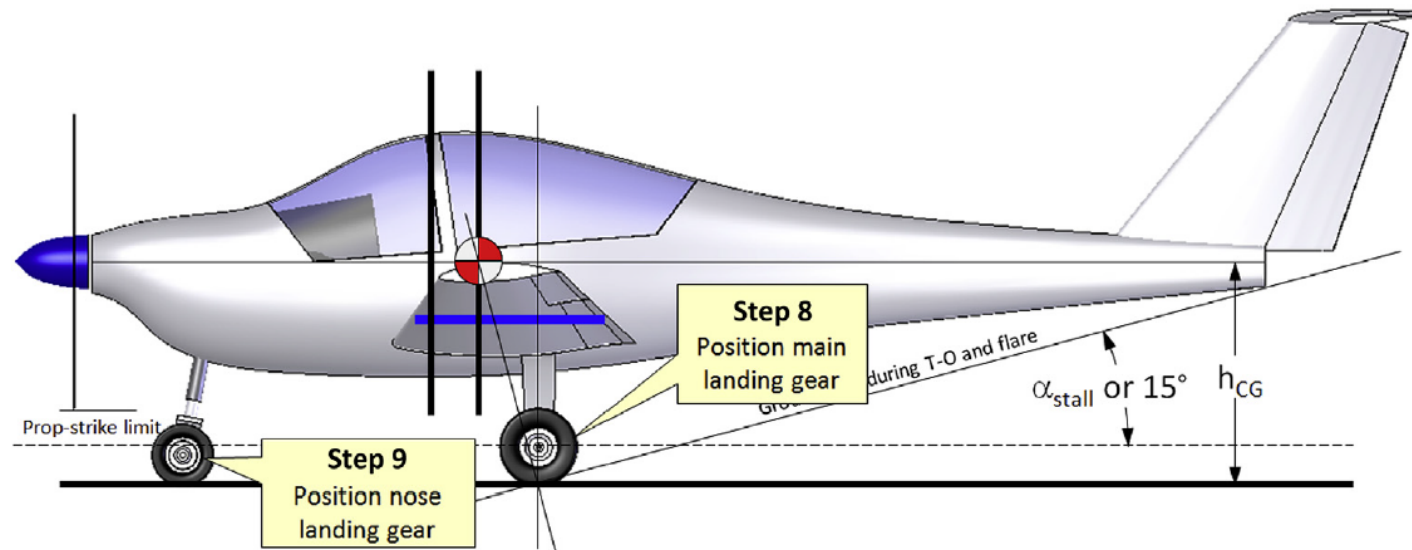
- ✈ Step 3: Determine the highest vertical CG location



(requires a complete CG envelope)

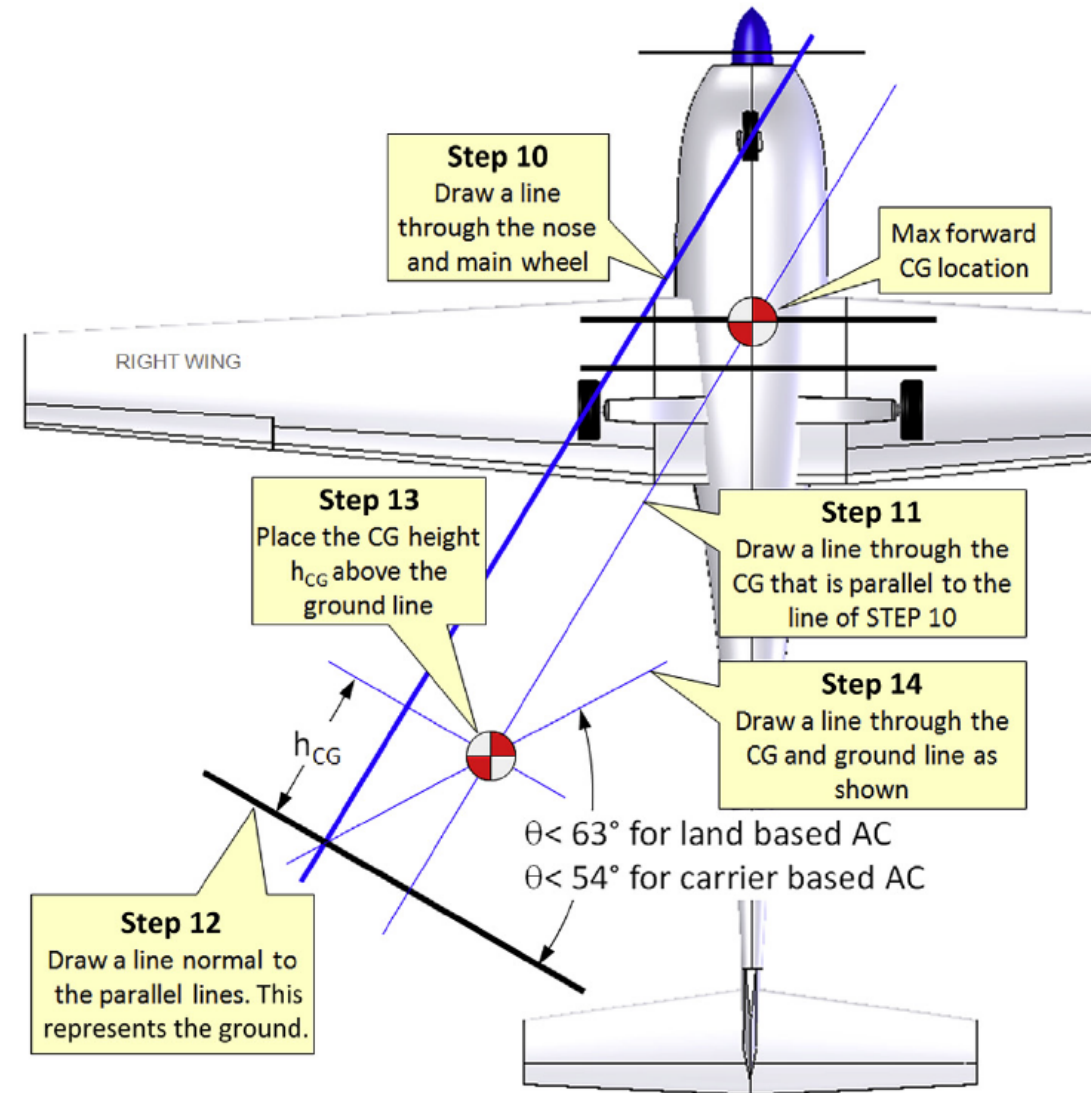


- ✈ Step 4: Draw the prop-strike limit as shown.
- ✈ Step 5: Draw the tail strike line (ensuring it is at the stall angle)
- ✈ Step 6: Draw a line through the CG perpendicular to the tail strike line => contact point of tire at static deflection of 1g load.



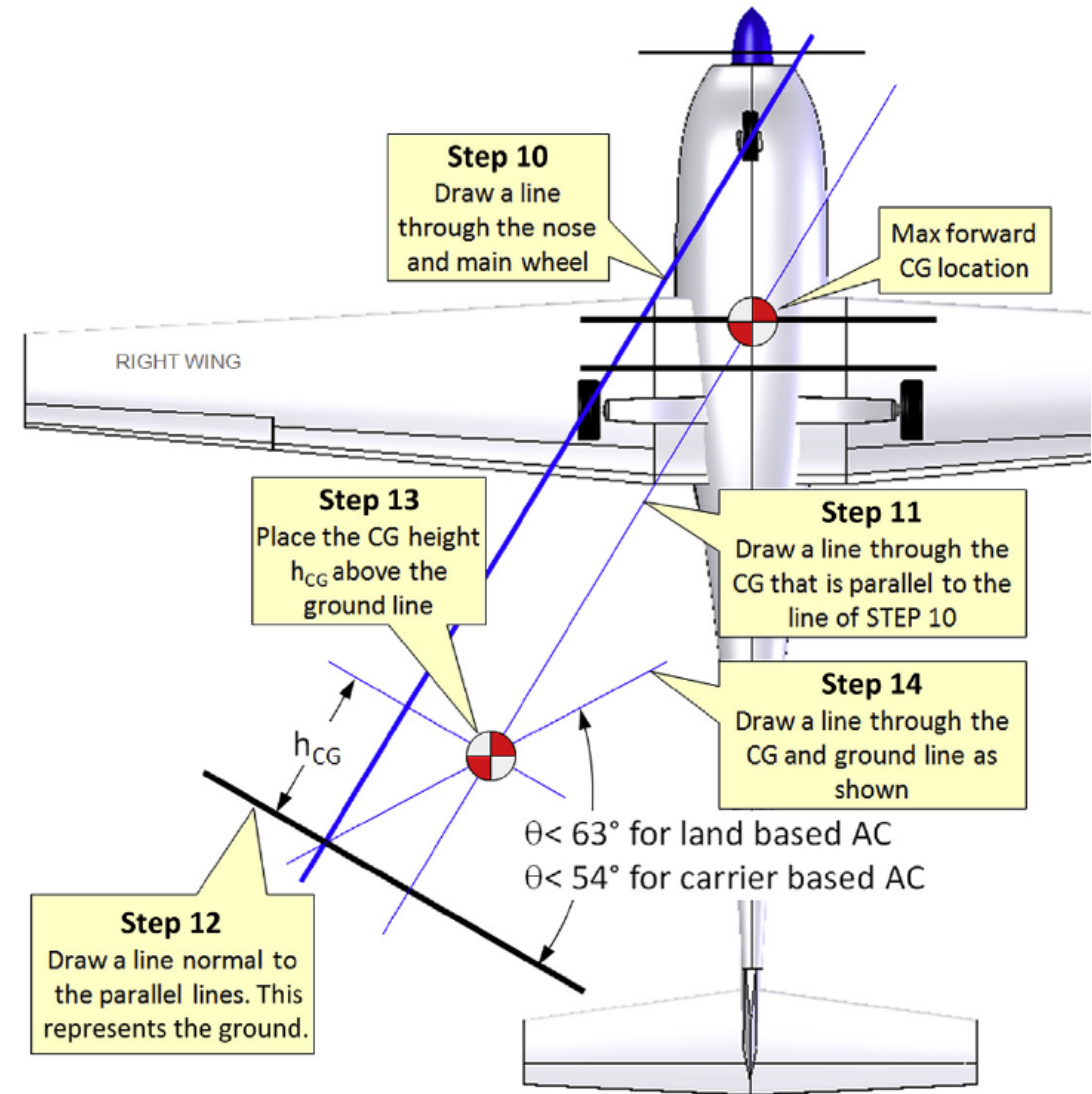
- ✈ Step 8: Position the main landing gear => contact point intersection point
- ✈ Step 9: Position the nose gear to carry:
 - No more than 20% of the weight when the CG is at the forward limit
 - No less than 10% of the weight when the CG is at aft limit

- ✈ Step 10: Draw a line through the center of the contact points.
- ✈ Step 11: Draw a line through the CG parallel to the line of step 10.
- ✈ Step 12: Draw a line ("ground") perpendicular to the lines drawn in 10 and 11.
- ✈ Step 13: place the CG distance h_{CG} above the ground line and along the line of step 11.



✈ Step 14: measure the overturn angle.
If larger then permitted:

- The wheel track can be increase.
- The height of landing gear can be increased.
- The spacing between nose and main landing gear can be increase.



Thank You!



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